EXCEPTIONAL CYCLES AND HARMONIC METRICS IN K3 BUNDLE MODULI

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Abstract. We construct compact moduli fourfolds of bundles on a K3 surface whose common contraction has a singular surface and one deeper point. In a specified stability chamber, a moving projective line specializes with multiplicity one to a line in each of two Lagrangian planes. We identify the primitive integral curve and surface classes and calculate their intersection matrix. Restriction from the complement to the boundary of a neighborhood of the planes gives an explicit congruence modulo eight determining their degree-four integral cohomological gluing. For the natural L^2 metric on moduli, defined by harmonic deformation representatives, these same cycles have exact areas and volumes and ambient diameter estimates uniform as the two positive stability parameters vanish at unrelated rates. At separated scales, rescalings based on the smaller plane converge in pointed Gromov–Hausdorff topology to the complete Calabi metric on $T^*\mathbb{P}^2$. The central harmonic metric completes to the common contraction, which is the compact Gromov–Hausdorff limit of the chamber spaces. The proof combines a complete boundary analysis with convergent symplectic coordinates and relative harmonic estimates. An isolated-plane contraction provides the corresponding congruence modulo three.

1. Introduction

A stability wall can contract a family of curves and a higher-dimensional exceptional fibre at the same time. The topology of the smooth moduli space then depends on how these cycles meet, while its metric degeneration depends on the relative rates at which their periods vanish. The two questions concern the same geometric objects. To relate them, a local description of a singularity must be placed inside a compact moduli problem, with its full boundary and its actual harmonic metric.

We carry this out for two explicit moduli problems on K3 surfaces. The first contracts one Lagrangian plane. The second has a moving family of projective lines and a deeper fibre consisting, in one chamber, of two Lagrangian planes meeting at a point. A moving line degenerates into one line in each plane. This specialization gives an integral relation among global curve classes and an exact identity among their harmonic areas. The intersection of the planes also changes the integral attachment: the relevant boundary group is cyclic of order eight, whereas the boundary of a single plane has order three.

1.1. The compact family.

For a sheaf on a K3 surface, use the Mukai vector and pairing

$$v(F) = (r, c_1, r + c_1^2/2 - c_2), \qquad (r, c, s) \cdot (r', c', s') = c \cdot c' - rs' - r's.$$

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Choose a K3 surface X with full Néron–Severi lattice generated by H, D_1, D_2, where H is ample and the intersection matrix is

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -6 & -4 \\ 0 & -4 & -8 \end{pmatrix}. \tag{1}$$

Such a surface exists by the construction in Section 4. Put

$$L_0 = \mathcal{O}_X, \quad L_1 = \mathcal{O}_X(D_1), \quad L_2 = \mathcal{O}_X(D_2), \qquad a_i = v(L_i), \qquad v = a_0 + a_1 + a_2.$$

Thus $v = (3, D_1+D_2, -4)$ and $v^2 = 2$. Let ω_η be the Ricci-flat representative of $H+sD_1+tD_2$, for $\eta = (s,t)$ near zero. The normalized stability parameters are

$$\zeta(\eta) = \frac{2\pi}{3} \begin{pmatrix} -10 & -12 \\ 8 & 0 \\ 2 & 12 \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix}, \qquad \zeta_0 + \zeta_1 + \zeta_2 = 0. \tag{2}$$

The three lines $\zeta_i = 0$ divide the parameter plane into six chambers. By a generic parameter below we mean a point outside these lines in a sufficiently small neighborhood of the origin.

On a smooth moduli space we use the real part of the unnormalized Dolbeault harmonic L^2 pairing, at fixed holomorphic determinant. Write g_η for this metric and Ω_η for its Kähler form. All distances between exceptional points are measured in the entire moduli space. Integral cohomology is understood unless another coefficient group is displayed.

Theorem 1.1. *For the family above, the following assertions hold.*

(1) *Each generic chamber σ has a smooth compact projective moduli fourfold M_σ of slope-stable bundles, of deformation type $X^{[2]}$. There are projective symplectic contractions $\pi_\sigma : M_\sigma \to Z$ to a common normal projective fourfold. The regular locus of Z is the moduli of H-stable bundles. Its reduced singular locus is a normal K3 surface T with one A_1 point o. The fibres over $T \setminus \{o\}$ are projective lines. Over o the fibre is two planes meeting transversely at a point in the chambers with signs $(+,-,+)$ and $(-,+,-)$; in the other four chambers it is $\mathbb{P}^2 \cup \mathbb{F}_1$, meeting along a line and the negative section. Every central or nearby ideal HYM object has zero lost charge, and every sheaf in these moduli problems is locally free.*

(2) *Fix $\zeta_0 = a > 0$, $\zeta_2 = b > 0$ and $\zeta_1 = -a - b$. Write P_0, P_2 for the two planes, c_0, c_2 for their line classes, and f for a moving fibre class. These classes satisfy*

$$f = c_0 + c_2, \qquad \left(\int_{M_\sigma} [P_i][P_j] \right)_{i,j=0,2} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}. \tag{3}$$

The curve and surface pairs each span a primitive rank-two integral subgroup. A proper flat analytic family realizes the curve relation with both multiplicities equal to one. For $V = M_\sigma \setminus (P_0 \cup P_2)$, let L be the boundary of a regular neighborhood of the planes. Then $H^4(L) = \mathbb{Z}/8$, and restriction to L is an explicit surjection $\kappa : H^4(V) \to \mathbb{Z}/8$ such that restriction identifies

$$H^4(M_\sigma) = \{(u, n_0, n_2) \in H^4(V) \oplus \mathbb{Z}^2 : n_0 - 3n_2 \equiv \kappa(u) \pmod 8\}. \tag{4}$$

Proposition 6.1 evaluates the map on Chern classes of family indices associated with a universal bundle. The complement cohomology ring, together with these distinguished restricted classes, is retained global data.

(3) *The same cycles have exact harmonic areas and volumes*

$$\begin{aligned} &\mathrm{Area}(c_0) = \pi a, \qquad \mathrm{Area}(c_2) = \pi b, \qquad \mathrm{Area}(f) = \pi(a+b), \\ &\mathrm{Vol}(P_0) = \frac{\pi^2 a^2}{2}, \qquad \mathrm{Vol}(P_2) = \frac{\pi^2 b^2}{2}. \end{aligned} \tag{5}$$

Their ambient diameters are comparable to $\sqrt{a}$ and $\sqrt{b}$, with constants independent of a/b. As $a/b \to 0$ and $a+b \to 0$, rescaling distances by $a^{-1/2}$ at P_0 gives the complete unit-level Calabi quotient on $T^*\mathbb{P}^2$, and

$$\frac{\mathrm{diam}_{M_\sigma} P_0}{\sqrt{a}} \longrightarrow \frac{\pi}{2}.$$

The reversed hierarchy has the corresponding conclusion for P_2. There are also complete successive pointed limits at any fixed finite number of scales, under the convergence hypotheses on normalized levels and normal basepoints in Theorem 8.1. The harmonic metric on Z_{reg} has compact completion (Z, d_0) with its projective topology, and

$$(M_{\sigma(\eta)}, g_\eta) \longrightarrow (Z, d_0) \quad \textit{in compact Gromov–Hausdorff distance}$$

along every generic sequence $\eta \to 0$.

The formula (4) records an integral additive attachment. Its order-eight congruence depends on the intersection of the two planes. It does not follow from the ranks of their cohomology groups. The complement V still contains the exceptional geometry over the moving stratum; it differs from the common stable open set. Section 7 describes that geometry and computes its ring over the open stratum, retaining the nonzero algebraic Brauer class of its projective-line bundle.

1.2. **A single plane and the proof mechanism.** Section 3 begins with the lattice $\mathrm{NS}(X) = \mathbb{Z}H \oplus \mathbb{Z}D$, $H^2 = 2$, $H \cdot D = 0$, $D^2 = -22$, and vector $(3, D, -4)$. The only strictly polystable central bundle is $\mathcal{O}_X \oplus B$, with B a rigid rank-two bundle. Each adjacent compact fourfold contracts a plane. Its primitive surface class has square three, and the boundary of a normal neighborhood has fourth cohomology $\mathbb{Z}/3$. The corresponding generator formula for the attachment is the one-component version of (4). This example also fixes the metric normalization: if $\kappa_t = 44\pi|t|/3$, an exceptional line has area $\pi\kappa_t$, and the plane has ambient diameter $(\pi/2)\sqrt{\kappa_t}(1 + O(|t|))$.

Three arguments connect these calculations to compact geometry. First, an exhaustive rank calculation uses the lower bound -2 for the Mukai square of a stable K3 sheaf to exclude every possible lost-charge vector. Surface gauge compactness then applies without concentration. Normalized inclusions and quotients in the central Jordan–Hölder filtrations show that entire exceptional fibres enter the local analytic charts. This supplies the global coverage needed before a quotient model can identify a contraction.

Second, universal extension families determine restrictions of integral classes to the exceptional cycles. These restrictions prove primitivity and evaluate the boundary homomorphism. The family $xy = t$ at the intersection of the two planes identifies the actual specialization multiplicities. The two ingredients serve different purposes: lattice pairings identify classes, while a proper flat family proves their geometric specialization.

Third, the harmonic comparison is relative in every quotient direction. Its error remains small even when one period is much smaller than the other. A lower bound for the length required to leave a chart then compares the two ambient distance infima. This explains why

the estimates apply to distances in the whole compact moduli space, and why the rescaled limits are complete. Global completion additionally uses the identification of the limiting harmonic class with the ample class of the projective contraction.

1.3. **Relation to earlier work.** The existence and deformation type of the compact moduli spaces are provided by Yoshioka's theorems [Yos00, Yos01]. We use the projective slope moduli of Greb–Toma [GT17] and the surface gauge comparison of Greb–Sibley–Toma–Wentworth [GSTW21]. Bayer–Macrì's theory [BM14] supplies the broader birational setting for stability walls. Local quiver descriptions of K3 sheaf moduli are developed by Arbarello–Saccà [AS18]. The two compact fibre configurations in Theorem 1.1 already occur in Bielawski–Dancer's hypertoric examples [BD00, Example 6.10]; here they are realized as the complete fibres of the stated compact slope-moduli contractions.

Formality is also an established input to this subject. Bandiera–Manetti–Meazzini [BMM21, Theorem 5.1] prove DG-Lie formality for coherent sheaves with linearly reductive automorphism group on a smooth projective surface with trivial canonical bundle. Their theorem applies to our common-slope direct sums. The related projectively hyperholomorphic formality theorem is [MO23, Theorem 4.3]. Section 2 gives the additional coordinate information used for the metric comparison: a convergent equivariant symplectic change whose first correction is cubic. Relative finite balancing and the actual harmonic reduction are the previously proved results [Wue26a, Wue26b], restated below with the required normalizations.

The plane-class identities and Hilbert-square cohomology are classical [HT09, Tot20, Göt90]; the calculation here identifies their integral representatives and attachments in the compact bundle-moduli problem. Markman's index generators [Mar07] make the attachment maps explicitly evaluable. For the global metric completion we use Song [Son14] and Tosatti [Tos09], after checking their geometric and metric hypotheses. The conclusions about cycle diameters and successive scales have separate local proofs.

The resulting theorems concern exceptional cycles of compact bundle-moduli contractions without bubbling. The complement and its distinguished integral classes supply the global information retained in the cohomological calculation.

2. Convergent harmonic symplectic models

We first explain how the local equations and the metric use the same coordinates. The qualitative quadraticity of the deformation problem is known. The point needed here is control of the coordinate change relative to the central harmonic norm.

2.1. **Conventions and the harmonic recursion.** Let (X, ω_0) be a Ricci-flat K3 surface and

$$E_0 = \bigoplus_{i=1}^{k} F_i \otimes W_i$$

a locally free slope-polystable bundle. The F_i are pairwise nonisomorphic stable bundles of common slope. Put $r_i = \operatorname{rank} F_i$ and $m_i = \dim W_i$, and choose a direct-sum Hermitian–Einstein metric h_0. Write

$$d = \bar\partial_{\operatorname{End} E_0}, \quad V = H^1(X, \operatorname{End} E_0), \quad G = \prod_i \mathrm{GL}(W_i), \quad K = \prod_i \mathrm{U}(W_i).$$

Ineffective factors are removed when taking effective quotients. Since $H^1(X, \mathcal{O}_X) = 0$, all first-order deformations preserve the holomorphic determinant. We use the h_0, ω_0 harmonic

norm. Degrees are $c_1(F)\cdot[\omega]$, with no division by the volume of X. The HE determinant connection varies with ω.

Let H_q and G_q be the harmonic projections and Green operators. At a fixed Sobolev order above the multiplication threshold, the analytic implicit-function theorem solves

$$a(x) = x - d^*G_2(a(x)\wedge a(x)), \qquad F(x) = H_2(a(x)\wedge a(x)) \tag{6}$$

for small $x \in V$. Thus $a(x) = x + O(|x|^2)$, and $F = 0$ is the full convergent Kuranishi obstruction. The operators commute with the parallel multiplicity matrices, so a and F are G-equivariant wherever both sides are defined. This is a local equivariance statement; a Euclidean ball is not G-invariant.

The surface hypothesis has a useful consequence. Since there are no $(0,3)$-forms, $I - H_2 = dd^*G_2$ on $(0,2)$-forms. Applying d to (6) gives the exact identity

$$da(x) + a(x)\wedge a(x) = F(x). \tag{7}$$

It holds before imposing integrability. We will use it to regard the obstruction itself as a moment map.

2.2. **A symplectic coordinate change with cubic correction.** Normalize the parallel holomorphic two-form σ_X compatibly with the harmonic norm, and define on $(0,1)$-forms

$$\mathcal{B}(\alpha,\beta) = \int_X \sigma_X \wedge \operatorname{tr}(\alpha\wedge\beta).$$

In an adapted unitary coframe the antilinear operator

$$J(A_1\bar\theta^1 + A_2\bar\theta^2) = -A_2^*\bar\theta^1 + A_1^*\bar\theta^2 \tag{8}$$

satisfies $J^2 = -1$ and $\mathcal{B}(\alpha, J\beta) = \langle\alpha,\beta\rangle_{L^2}$. It preserves harmonic forms. Indeed the common-slope HE equation makes the induced connection on $\operatorname{End} E_0$ anti-self-dual; the Kähler identities, or integration by parts in the displayed pairing followed by $J^2 = -1$, give closedness and coclosedness of $J\beta$ when β is harmonic. This is the usual quaternionic structure in the ASD moduli construction [Ito88].

Let $B = a^*\mathcal{B}$ on V. It is closed and holomorphic. Pairing curvature with a parallel endomorphism identifies its complex moment with F, by (7) and Stokes' theorem. For $H_\xi(a) = \int_X \sigma_X \wedge \operatorname{tr}(\xi a\wedge a)$, graded cyclicity of the trace gives

$$dH_\xi(a)(b) = \int_X \sigma_X \wedge \operatorname{tr}\big(\xi(b\wedge a + a\wedge b)\big) = \mathcal{B}([\xi,a],b).$$

We use this convention for both the linear and nonlinear complex moments.

The form B_0 is the Serre pairing on V. The correction $Da_x - I$ takes values in $\operatorname{im} d^*$ and is $O(|x|)$. A harmonic form is $\mathcal{B}$-orthogonal to $\operatorname{im} d^*$, by (8) and ordinary harmonic orthogonality. The two terms linear in $Da_x - I$ therefore vanish, leaving

$$B_x = B_0 + O(|x|^2). \tag{9}$$

Proposition 2.1. *There is a K-equivariant holomorphic coordinate change on a neighborhood of zero in V such that*

$$\Psi^*B = B_0, \qquad \Psi(x) = x + O(|x|^3), \qquad D\Psi_x = I + O(|x|^2). \tag{10}$$

Under the fixed Serre identification of the obstruction target with the dual parallel algebra,

$$F\circ\Psi = Q,$$

where Q is the quadratic complex moment of the linear action on (V, B_0). The equality identifies the full convergent obstruction ideals. Stable-factor ranks and multiplicities are arbitrary.

Proof. Write $\Delta B = B - B_0$. The holomorphic radial homotopy supplies a one-form α with $d\alpha = \Delta B$. In coordinates,

$$\alpha_x(v) = \int_0^1 t\, \Delta B_{tx}(x, v)\, dt.$$

Thus $\alpha = O(|x|^3)$, and it is K-invariant. On one smaller ball all forms $B_s = B_0 + s\Delta B$, $0 \le s \le 1$, are nondegenerate. Solve $\iota_{V_s} B_s = -\alpha$. The resulting holomorphic vector fields are invariant and vanish to order three, uniformly in s. Their flows exist through time one on a common smaller neighborhood. Cartan's formula gives

$$\frac{d}{ds} \Psi_s^* B_s = \Psi_s^* (\Delta B + d\iota_{V_s} B_s) = 0.$$

Their order of vanishing gives the two estimates in (10). Holomorphic K-equivariance extends locally to G-equivariance.

The two maps $F \circ \Psi$ and Q are moment maps for the same linear action and the same form B_0. Pairing their difference with every parallel endomorphism has zero differential. Both vanish at zero, so their difference is zero. The statement is an equality of analytic maps, hence of their generated ideals. □

Paired harmonic orthonormal bases identify the representation with a doubled Ext quiver, simultaneously normalizing its norm and complex pairing. To track ranks, let η_X be the harmonic $(0,2)$-form with $\int_X \sigma_X \wedge \eta_X = 1$. The harmonic representative of a multiplicity-trace covector (q_i) is

$$\bigoplus_i r_i^{-1} I_{F_i} \otimes q_i \eta_X.$$

These factors belong to the obstruction-target identification. The real moment is paired with the ordinary multiplicity trace. For a nearby polarization parameter η its target is

$$\zeta_i(\eta) = 2\pi \big(r_i \mu_{\omega_\eta}(E_0) - \deg_{\omega_\eta} F_i \big), \qquad \sum_i m_i \zeta_i(\eta) = 0. \tag{11}$$

A determinant-preserving lift subtracts a rank-weighted common scalar, which acts trivially on the deformation representation.

The DG-Lie formality theorem of [BMM21, Theorem 5.1] already applies to E_0, because $\mathrm{Aut}(E_0)$ is reductive. Proposition 2.1 specifies convergent symplectic coordinates and their derivative, which are the data required in the next comparison. Equal reduced Hilbert polynomials are not part of its hypotheses.

2.3. **The actual harmonic metric.** The full local HYM theorem [Wue26b, Theorem 1.1 and Sections 3,5] applies to the convergent zero locus $F^{-1}(0)$. It gives nested coefficient neighborhoods and an open classifying homeomorphism to the actual fixed-determinant HYM moduli, smooth on smooth quotient pieces and covering every sufficiently nearby solution. At a balanced coefficient x, its comparison form is

$$g_{\mathrm{fin}}([b],[b]) = \inf_{\xi \in \mathrm{Lie}\, G} \|b - C_x \xi\|^2, \qquad C_x \xi = [\xi, x]. \tag{12}$$

The finite orbit operator is $C_x^* C_x$; its positive eigenvalues measure the nonzero infinitesimal orbit directions. The actual harmonic metric lies between $1 - C(|\eta| + |x|^2)$ and $1 + C(|\eta| + |x|^2)$ times this form. The comparison concerns tangents to actual smooth quotient pieces, not all

Zariski tangent vectors at a singular point. Constants depend on the fixed central background, its complementary elliptic gaps and bounded smooth parameter data. They do not depend on ratios of positive eigenvalues of $C_x^*C_x$ or on the nearby stabilizer type.

We record precisely how the coordinate change preserves this order. Pull back the flat real Kähler form by Ψ. By (10) its error is $O(r^2)$ and its moment is $\mu \circ \Psi = \mu + O(r^4)$, where $r = |x|$. These forms satisfy the exact Hamiltonian identity. The finite balancing theorem [Wue26a, Theorem 2.1], applied on balls of radius comparable to r, changes the real orbit by e^S, $\|S\| = O(r^2)$, to put the pulled-back and flat moments at the same central target. The theorem includes the inverse correspondence and permits the invariant analytic subset $Q^{-1}(0)$.

For every ξ, equivariance gives $D\Psi_x C_x\xi = C_{\Psi(x)}\xi$ and

$$(1 - Cr^2)\|b - C_x\xi\|^2 \le \|D\Psi_x(b - C_x\xi)\|^2 \le (1 + Cr^2)\|b - C_x\xi\|^2.$$

Infimizing preserves the inequalities. The derivative of the varying balancing gauge contributes a complex orbit vector, which disappears in this same infimum; its value has norm distortion $1 + O(r^2)$. Thus a derivative estimate for that gauge, which need not be uniform at a stabilizer change, is unnecessary.

Corollary 2.2. *Put $\mathcal{Q}_\zeta = \{Q = 0, \mu = \zeta\}/K$, with its flat quotient metric. On fixed nested neighborhoods there are classifying identifications Φ_η with the actual HYM moduli satisfying*

$$(1 - C(|\eta| + r^2))g^{\mathrm{quot}}_{\zeta(\eta)} \le \Phi_\eta^* g_\eta \le (1 + C(|\eta| + r^2))g^{\mathrm{quot}}_{\zeta(\eta)} \tag{13}$$

on every smooth quotient piece. Their images cover every sufficiently nearby solution. The constants are independent of positive finite scale ratios. The fixed holomorphic family $x \mapsto (E_0, \bar\partial_0 + a(\Psi(x)))$ determines the classifying maps; the HE gauge corrections preserve its isomorphism classes.

On smooth complex quotient charts the classifying maps are holomorphic. Their open injectivity makes them biholomorphisms onto their images. Statements about entire exceptional fibres will require the additional compactness arguments below; they are not consequences of a germ identification alone.

2.4. **Mukai and harmonic normalizations.** For a universal family $\mathcal{E}$ on $X \times M$, fix

$$\theta_v(x) = \Big[p_{M*}\big(\mathrm{ch}(\mathcal{E})\sqrt{\mathrm{td}_X}\, p_X^*(x^\vee)\big)\Big]_2, \qquad x^\vee = (r, -c, s), \quad x \in v^\perp. \tag{14}$$

The perpendicular is taken in the full integral Mukai lattice. Twisting $\mathcal{E}$ by a line from M does not change this map on $v^\perp$. We also use its real extension and the Beauville–Bogomolov form q, including the rational dual form on curve functionals.

The unnormalized real Dolbeault metric is half the usual real connection L^2 metric. In this convention its Kähler class is

$$[\Omega_\eta] = 2\pi^2\theta_v\left(0, [\omega_\eta], \frac{c_1(v)\cdot[\omega_\eta]}{\mathrm{rank}\, v}\right). \tag{15}$$

To verify the factor, set $\Delta(\mathcal{E}) = c_2(\mathcal{E}) - (r-1)c_1(\mathcal{E})^2/(2r)$. Biswas–Schumacher's fixed-determinant formula [BS16, Proposition 4.3, (4.6)] is

$$\frac{\omega^{\mathrm{WP}}}{2\pi^2} = -\frac{1}{r}p_{M*}\big(\mathrm{ch}_2(\mathrm{End}\,\mathcal{E})\omega_\eta\big) = 2p_{M*}\big(\Delta(\mathcal{E})\omega_\eta\big).$$

Their form is $iG_{j\bar k}\, dz_j \wedge d\bar z_k$, whereas the real metric $\mathrm{Re}\, G$ has Kähler form $\omega^{\mathrm{WP}}/2$. Expanding (14) identifies the last pushforward with the Mukai expression in (15). This also shows that

base-line twists cancel. In particular cohomological periods and quotient moment scales can now be compared without an unspecified normalization constant.

3. A compact contraction of one plane

The isolated example separates the two ingredients of the construction. The Mukai square bound controls the entire compactification, while the three-dimensional extension space identifies the exceptional plane. Its topology and metric can then be calculated in the same normalization.

3.1. **The surface and the central factors.** Take the even lattice

$$\mathrm{NS}(X) = \mathbb{Z}H \oplus \mathbb{Z}D, \qquad H^2 = 2, \quad H \cdot D = 0, \quad D^2 = -22.$$

Morrison's realization theorem [Mor84, Corollary 2.9(i)] gives a projective K3 surface with precisely this Néron–Severi lattice. A root would satisfy $x^2 - 11y^2 = -1$, which is impossible modulo eleven. The positive cone is therefore an ample chamber; choose its component so that H is ample. Put

$$a = (1, 0, 1), \qquad b = (2, D, -5), \qquad v = a + b = (3, D, -4). \tag{16}$$

We have $a^2 = b^2 = -2$, $(a, b) = 3$, $v^2 = 2$ and $c_2(b) = c_2(v) = -4$.

Lemma 3.1. *There is a unique H-slope-stable bundle B with vector b. It is rigid. Every central H-polystable class of vector v is either stable or $\mathcal{O}_X \oplus B$. Every H-semistable torsion-free sheaf of vector v is locally free. No central ideal object with that total vector has positive lost charge.*

Proof. Yoshioka's primitive-vector theorem [Yos00, Theorem 0.1 and condition (♮)] gives a sheaf of vector b at a generic polarization arbitrarily near H. Choose the polarization in a chamber adjoining H for the finite slope-wall arrangement [GT17, Theorem 6.6]. Generic slope equality would require $D/2$ to be integral. It is not, so the sheaf is slope stable. As the polarization tends to H within that chamber, every fixed subsheaf inequality becomes weak; the sheaf is H-semistable.

An H-slope-zero rank-one torsion-free factor has the form $\mathcal{O}_X(kD) \otimes I_T$, with vector $(1, kD, 1 - 11k^2 - n)$, where $n = \operatorname{length} T \geq 0$. If a sheaf of vector b split into two such factors, its second Chern class would be

$$22k(k-1) + n_1 + n_2 \geq 0,$$

contrary to $c_2(b) = -4$. Thus it is H-stable.

For any slope-stable torsion-free sheaf F, its double dual is stable: intersect a subsheaf of F^{**} with F to preserve rank and first Chern class. If $l = \operatorname{length}(F^{**}/F)$, then

$$v(F^{**})^2 = v(F)^2 - 2\operatorname{rank}(F)l. \tag{17}$$

A simple sheaf on a K3 surface has Mukai square at least -2, by Riemann–Roch and Serre duality. For $b^2 = -2$ this forces $l = 0$. Two nonisomorphic stable bundles with vector b would have $\mathrm{Hom} = \mathrm{Ext}^2 = 0$, whereas their Euler pairing is two. This is impossible. The same calculation for B itself gives $\mathrm{Ext}^1(B, B) = 0$.

Now allow an ideal object with residual vector $v_l = (3, D, -4 + l)$, where $l \geq 0$ is its total lost charge. The residual stable factors have slope zero. A stable rank-three factor has square $2 - 6l \geq -2$, hence $l = 0$. For a rank partition $1 + 2$, take the rank-one vector $c = (1, kD, 1 - 11k^2 - n)$. The complementary stable rank-two vector has square

$$(v_l - c)^2 = -2 + 44k - 66k^2 - 4(n + l). \tag{18}$$

For integral k, the quadratic term is nonpositive, with equality only at zero. The lower square bound forces $k = n = l = 0$. The factors are $\mathcal{O}_X$ and B.

For three rank-one factors, write $\sum k_i = 1$. Their second Chern class is

$$11\left(\sum_i k_i^2 - 1\right) + \sum_i n_i \geq 0,$$

whereas $c_2(v_l) = -4 - l < 0$. These are all positive-rank partitions. Finally a stable sheaf of vector v is locally free by (17), and a strictly semistable one is an extension of the two locally free factors. Such an extension is locally free. □

We will also need the following consequence of surface gauge compactness, which applies to both examples in this paper.

Lemma 3.2. *Fix a topological bundle type on a K3 surface and a smooth compact family of base Kähler metrics through ω_0. Keep the holomorphic determinant fixed and let its HE connection vary smoothly. Suppose every central ideal HYM object of that total type has zero lost charge. A sequence of HYM connections whose parameters tend to zero then has a subsequence converging smoothly modulo gauge on all of X. Moreover, after shrinking the parameter neighborhood, no ideal object of that total type has positive lost charge at any parameter in it.*

Proof. The trace-free connections are projective ASD connections, and their energies are topologically bounded. Pass to a weak limit of their curvature measures and remove the finitely many atoms above the small-energy threshold. In fixed coordinate balls away from those points, the base metrics converge smoothly. Their Sobolev, gauge-fixing and elliptic constants are uniformly bounded. Small-energy regularity first gives local curvature bounds stronger than L^2; varying-metric compactness and elliptic bootstrapping then give smooth subsequential convergence in gauges on the complement. This is the setting of [Weh04, Introduction, Theorem E′, and Theorem 10.3]. An energy bound alone would not meet that theorem's stronger local integrability hypothesis.

The surface compactness, extension, removal and defect-current results [GSTW21, Section 3.1, Theorems 3.3, 3.5–3.7 and 3.10] produce a central polystable residual bundle and a nonnegative integral lost charge. The determinant is smooth and has no concentrating part. Local projective instantons lift over four-balls, so the missing charges are integers. By hypothesis their sum is zero. The energy identity and small-energy regularity therefore remove all the exceptional points, proving smooth convergence on X.

If positive-charge ideal objects occurred arbitrarily close to zero, choose such a sequence. The energy bound permits a subsequence with one fixed positive defect. Applying compactness to its residual connections can add only nonnegative lost charge. The resulting central ideal object would still have positive defect, a contradiction. This also accounts for the zero-dimensional discrepancy of a torsion-free sheaf from its polystable double dual. □

3.2. **The two chambers and their full exceptional planes.** Let ω_t be the Ricci-flat representative of $H + tD$. The base volume is $1 - 11t^2$. Shrink the interval so that no other slope wall for b or v meets it. For large integral m, the degree of v at $mH \pm D$ is ∓ 22, coprime to three. Thus its Gieseker semistable sheaves are slope stable. Yoshioka's theorem and its deformation-type refinement [Yos01, Appendix, Theorem 8.1] give smooth projective fourfolds

$$M_\pm = M_{mH\pm D}(v)$$

of $\mathrm{K3}^{[2]}$ type. All their points are bundles, by (17). Since $\chi(E) = -1$, the universal-family criterion [HL10, Theorem 4.6.5] gives ordinary universal families.

Every chamber object is H-semistable, by taking limits of its subsheaf inequalities; an H-stable object remains stable in both adjacent chambers. The only remaining central graded type is $\mathcal{O}_X \oplus B$. Cross Hom and second Ext groups vanish, and both cross first Ext groups have dimension three. At $t > 0$ the allowable orientation is

$$0 \longrightarrow B \longrightarrow E \longrightarrow \mathcal{O}_X \longrightarrow 0, \tag{19}$$

since $\mu_t(B) = -11t < \mu_t(E) = -22t/3$. The orientation is reversed at $t < 0$.

The full local model at $E_0 = \mathcal{O}_X \oplus B$ has coordinates $(u, \varphi) \in \mathbb{C}^3 \oplus (\mathbb{C}^3)^*$, with action $(u, \varphi) \mapsto (e^{i\theta}u, e^{-i\theta}\varphi)$, complex equation $\varphi(u) = 0$, and real level

$$|u|^2 - |\varphi|^2 = \tau(t), \qquad \tau(t) = -\frac{44\pi}{3}t. \tag{20}$$

Here u is in $\mathrm{Ext}^1(B, \mathcal{O}_X)$. Formula (11) gives the sign and coefficient. The determinant-one infinitesimal lift has weights $2/3, -1/3$, whose difference is one; there is no additional rank factor in the effective circle moment.

For $\tau > 0$ the complex quotient is $T^*\mathbb{P}^2$ with coordinate $[u]$ on the zero section; for $\tau < 0$ use $[\varphi]$. The affine map is

$$(u, \varphi) \longmapsto u \otimes \varphi, \qquad \{A \in \mathfrak{sl}_3 : A^2 = 0,\ \operatorname{rank} A \leq 1\}. \tag{21}$$

The entire zero section has coefficient norm squared $|\tau|$. It therefore lies in the common coefficient neighborhood for small t. Corollary 2.2 gives actual stable bundles on that entire section. The coordinate change fixes each one-direction extension axis: its equivariance and identity derivative exclude higher monomials of weight one on that axis. The harmonic recursion is also linear there, because the square of a one-direction extension is zero. Thus these are the actual projective extension families in (19) and its reverse.

For clarity, the same chart contains every point of the actual central fibre. If not, take $t_n \to 0$ and bundles outside it with graded class $\mathcal{O}_X \oplus B$. Lemmas 3.1 and 3.2 give a smooth gauge limit G. Normalize the inclusion of the fixed first factor in each extension to have L^2 norm one. Its holomorphic-map equation has smoothly convergent coefficients; global elliptic estimates give smooth convergence to a nonzero map into G. Equal-slope stability and polystability split off that factor. The remaining factor is identified by its vector and the uniqueness just proved. Thus $G = \mathcal{O}_X \oplus B$, which puts the sequence in the open actual solution chart, a contradiction. The extension planes are embedded: the classifying map is biholomorphic on the smooth quotient, and the middle term determines its distinguished subobject up to scalar. Write their images as $P_\pm$.

3.3. **The common normal target.** The smooth chamber bases are weakly normal, so the Greb–Toma classifying theorem applies to their universal families [GT17, Main Theorem]. For a surface the slope equivalence is given by the graded double dual and its defect cycle [GT17, Theorem 5.5, Proposition 5.8 and Remark 5.9]. Our maps have a common reduced projective image: its points are the common stable open set and the one class $z_0 = \mathcal{O}_X \oplus B$. Their fibres are points and the entire planes just identified.

Normalize this reduced image. Stein factorization of either proper birational map has a normal finite factor. Its finite fibres have one point because the original fibres are connected. It is therefore the normalization, bijective over the reduced image. The two chamber maps lift to a common normal projective target Z, with the same point topology and the same fibres.

We explain the analytic identification, since a topological description of the fibres would not suffice. The local holomorphic classifying family commutes with central S-equivalence, which in the finite model is exactly the invariant $u\otimes\varphi$. Its chart contains the full exceptional plane. Properness now allows the target to be shrunk until its entire inverse image is in that chart: otherwise a sequence outside the chart mapping to z_0 would have a limit in the plane. The proper incidence map in (21) has the same property.

Both targets are normal. For the finite target, the hypersurface $\varphi(u)=0$ is an irreducible quadric with an isolated singularity, hence normal, and its invariant ring is normal. A proper birational map to a normal target has direct image of its structure sheaf equal to that structure sheaf. Pull back target coordinate functions to the two identified resolution neighborhoods and descend them by this identity. The resulting maps between target germs are inverse on the dense stable open set, hence everywhere. This identifies the actual contraction with (21).

Consequently z_0 is the unique singular point and both maps are small symplectic resolutions. Zero-level points with $u,\varphi\neq 0$ and $\varphi(u)=0$ are stable and tend to the origin. The listed target is therefore the closure of the stable wall locus. The surface comparison [GSTW21, Theorem 4.11 and Corollary 4.15] identifies its compact topology with the HYM compactification; its analytic germ was proved separately above.

At the central polarization, the ordinary Gieseker moduli is a different object. Its reduced Euler characteristic is $-1/3$, which cannot equal that of a factor of rank one or two. The nonsplit extensions in (19) are Gieseker stable at H, since the B subobject has reduced constant term $-3/2$; the opposite orientation is destabilized by $\mathcal{O}_X$. Thus $M_H(v)$ is the resolution M_+, not Z. The distinction between the slope target and the central Gieseker space is already visible in this simplest example.

The construction has now placed the full plane, with its cotangent normal bundle and its normalized quotient metric, inside a compact fourfold. Sections 6 and 8 compute its integral attachment and metric collapse. They will use the same extension family and the same signed Mukai marking.

4. The compact family with interacting strata

We now construct the family of Theorem 1.1. Its central factor list is short, but one factor varies in a K3 surface. The degeneration of that surface accounts for the interaction at the deeper point.

4.1. **Lattice realization and the polarization plane.** The lattice (1) has signature $(1,2)$ and determinant 64. In three hyperbolic planes with $e_i\cdot f_i=1$, it has the primitive embedding

$$H=e_1+f_1,\qquad D_1=e_2-3f_2,\qquad D_2=-4f_2+e_3-4f_3. \tag{22}$$

Projection to the three e-coordinates restricts to the identity on these generators, proving primitivity. Morrison's theorem [Mor84, Corollary 2.9(i)] realizes the lattice as the full Néron–Severi group of a projective K3 surface. A root would solve

$$x^2-3y^2-4yz-4z^2=-1.$$

Modulo four, even y would give $x^2=3$, and odd y would give $x^2=2$. There are no roots. Choose the positive component so that H is ample.

For the line bundles L_i introduced above, the Mukai data are

$$\begin{aligned} &a_0 = (1,0,1), \quad a_1 = (1, D_1, -2), \quad a_2 = (1, D_2, -3), \\ &v = (3, D_1 + D_2, -4), \qquad w = a_0 + a_2 = (2, D_2, -2), \\ &((a_i, a_j))_{i,j=0}^2 = \begin{pmatrix} -2 & 1 & 2 \\ 1 & -2 & 1 \\ 2 & 1 & -2 \end{pmatrix}. \end{aligned} \tag{23}$$

The pair sums a_0+a_1 and a_1+a_2 are spherical, whereas w is primitive isotropic and $(a_1, w) = 2$. Every nonzero difference of the line Chern classes has H-degree zero. Neither it nor its negative is effective, so cross Hom and second Ext groups vanish. The three edge dimensions of the doubled Ext quiver are therefore $1, 2, 1$.

For $\eta = (s, t)$ the Ricci-flat form has class $H + sD_1 + tD_2$ and volume $1 - 3s^2 - 4st - 4t^2$. The degrees of L_0, L_1, L_2 are $0, -6s - 4t, -4s - 8t$. Substitution in (11) gives (2), a rank-two linear map. The equations $\zeta_i = 0$ will be shown to be all the nearby actual walls.

4.2. **An exhaustive central factor calculation.** Define

$$Q(k, j) = 6k^2 + 8kj + 8j^2 = 4k^2 + 8(j + k/2)^2.$$

An H-slope-zero rank-one torsion-free sheaf has vector $c = (1, kD_1 + jD_2, 1 - Q(k, j)/2 - n)$, where $n \geq 0$.

Lemma 4.1. *The H-polystable classes of total vector v are precisely the stable classes, the classes $L_1 \oplus F$ with F H-stable of vector w, and $E_0 = L_0 \oplus L_1 \oplus L_2$. For vector w the only strictly polystable class is $L_0 \oplus L_2$. All the semistable sheaves in these two problems are locally free, and no central ideal object has positive lost charge.*

Proof. For residual vector $v_l = (3, D_1 + D_2, -4 + l)$, a stable rank-three factor has square $2 - 6l \geq -2$, so $l = 0$. For a rank partition $1 + 2$, the complementary square is

$$(v_l - c)^2 = -2 + F(k, j) - 4(n + l), \qquad F(k, j) = 20k + 24j - 18k^2 - 24kj - 24j^2. \tag{24}$$

Completing the square gives

$$F(k, j) = \frac{22}{3} - 3Q(k - 1/3, j - 1/3).$$

The inequality $F \geq 0$ implies $|k - 1/3|^2 \leq 11/18$, hence $k = 0, 1$. For $k = 0$ it gives $F = 24j(1 - j)$ and $j = 0, 1$; for $k = 1$ it gives $F = 2 - 24j^2$ and $j = 0$. The three values are $0, 0, 2$, all smaller than four, so $n = l = 0$.

The complementary vectors for $(k, j) = (0, 0), (0, 1)$ are a_1+a_2 and a_0+a_1. Neither admits an H-stable sheaf. Indeed Hom in both directions between such a stable rank-two sheaf and the corresponding sum of the two distinct stable lines vanishes. Serre duality then makes their Euler characteristic nonpositive, contradicting the value two. The remaining possibility is $L_1 \oplus F$ with $v(F) = w$.

For three rank-one factors let $\sum(k_i, j_i) = (1, 1)$ and let N be the sum of their singularity lengths. The second Chern identity is

$$\sum_i Q(k_i, j_i) + 2(N + l) = 14. \tag{25}$$

The nonzero lattice vectors of $Q \leq 14$ are

$$\begin{array}{l|l} Q = 6 & (1, 0), (1, -1), (-1, 0), (-1, 1) \\ Q = 8 & (0, 1), (0, -1). \end{array}$$

This list follows directly from $Q = 4k^2 + 8(j + k/2)^2$. At most two nonzero summands fit the budget. A single summand $(1, 1)$ has $Q = 22$. Two $Q = 6$ vectors cannot sum to $(1, 1)$, and the only $Q = 6$ plus $Q = 8$ pair with that sum is $(1, 0) + (0, 1)$. Thus (25) forces the unordered triple $\{0, D_1, D_2\}$ and $N = l = 0$.

For the auxiliary vector $w_l = (2, D_2, -2 + l)$, stability gives $-4l \geq -2$, hence $l = 0$. A split rank-one pair must satisfy

$$Q(k, j) - 4k - 8j + n + n' + l = 0.$$

Its quadratic part is $Q(k, j - 1/2) - 2$. The expression $4k^2 + 8(j - 1/2 + k/2)^2 \leq 2$ forces $k = 0$ and $j = 0, 1$, and then $n = n' = l = 0$. The pair is L_0, L_2. Stable sheaves of vectors w and v are locally free by (17). All semistable factors just listed are therefore bundles, and so are their extensions. □

Applying Lemma 3.2 to v and w excludes ideal charge uniformly in a smaller parameter neighborhood. This step includes parameters on the walls. The compact spaces constructed next have no additional singular-sheaf boundary.

4.3. **The finite model and its affine strata.** Use coordinates $(z, z', w, w', u, \varphi)$, with $u \in \mathbb{C}^2$ and $\varphi \in (\mathbb{C}^2)^*$. The single arrows are $z : 1 \to 0$ and $w : 1 \to 2$, and the double arrow is $u : 0 \to 2$; primes and φ denote the reverse arrows. The effective compact group is $K = (\mathrm{U}(1))^3/\mathrm{U}(1)$. By Proposition 2.1, the full complex equations and real moments are

$$\begin{gathered} zz' - \varphi(u) = 0, \qquad ww' + \varphi(u) = 0, \\ \mu_0 = |z|^2 - |z'|^2 - |u|^2 + |\varphi|^2, \qquad \mu_2 = |w|^2 - |w'|^2 + |u|^2 - |\varphi|^2, \\ \mu_1 = -\mu_0 - \mu_2. \end{gathered} \tag{26}$$

Write $\mathcal{Q}_\zeta = \{\text{complex equations}, \mu = \zeta\}/K$.

If the nonzero edge support were disconnected, summing the real moment over a connected component would force a nonempty proper subset sum of ζ to vanish. For generic ζ this is impossible. Connected support has trivial effective stabilizer, since all vertex dimensions are one. Thus the generic quotients are smooth. Torus GIT makes their maps to $\mathcal{Q}_0$ projective; arbitrary real parameters in a chamber give the same complex quotient as a rational parameter.

The affine zero space $Y \subset \mathbb{C}^8$ is normal. Its two equations are a regular sequence, so it is a pure six-dimensional complete intersection. Over $p = \varphi(u) \neq 0$ it is irreducible, with fibre $(\mathbb{C}^*)^2$ over an irreducible open subset of $\mathbb{C}^4$. The part $p = 0$ has dimension five, so is not another component. For a nonzero linear combination of the Jacobian rows to vanish, coefficients λ, ν must satisfy

$$\lambda z = \lambda z' = \nu w = \nu w' = 0, \qquad (\nu - \lambda)u = (\nu - \lambda)\varphi = 0.$$

The largest such locus is $z = z' = w = w' = 0$, $\varphi(u) = 0$, of dimension three. Thus Y is regular in codimension one and normal. Its torus invariant ring is normal as well.

A zero-level representation with

$$z = z' = w = 1, \quad w' = -1, \quad u = (1, 0), \quad \varphi = (1, 0) \tag{27}$$

is simple as a representation of the quiver with these relations, or *module-simple*: its directed support is strongly connected. It gives a nonempty stable open set, so $\mathcal{Q}_\zeta \to \mathcal{Q}_0$ is birational and has connected fibres. Among the proper connected dimension vectors, only the double-edge pair $(1, 0, 1)$ supports a nonrigid central simple factor. Its affine quotient is the A_1 surface. A single-edge pair has equation $xy = 0$ and at zero real level cannot have connected

support. Hence the affine quotient has precisely its smooth four-dimensional stratum, the smooth part of that A_1 surface, and the vertex.

4.4. **Compact moduli and whole fibres.** Yoshioka's theorems give generic-polarization moduli of vectors v and w, of K3$^{[2]}$ and K3 type. Generic Gieseker stability is slope stability because $(D_1 + D_2)/3$ and $D_2/2$ cannot be normalized Chern classes of proper positive-rank subsheaves. The finite slope-wall theorem supplies such polarizations arbitrarily near H. Lemma 4.1 makes every object a bundle. For v, $\chi(v) = -1$ gives an ordinary universal family. For w, the gcd of the algebraic Mukai pairings is two. Căldăraru's theorems [Cal02, Theorems 1.2(1), 3.2 and 3.4] give an obstruction of exact order two and the integral isotropic Mukai isometry. We use local universal w-families and their coarse descent.

Every chamber bundle is H-semistable. At the deepest class E_0 the chart (26) contains the entire actual fibre for sufficiently small parameters. To prove this, take any sequence in that fibre. No-charge compactness gives a smooth gauge limit G. Pass to a subsequence on which the first and last factors of its central Jordan–Hölder filtration are fixed lines L_i, L_j; they are distinct because the three factors occur once. Normalize the inclusion $L_i \to E_n$ and quotient $E_n \to L_j$ separately in L^2, using the same gauges as for the connections. Elliptic estimates give nonzero smooth limiting maps. Polystability splits off these two distinct lines from G. The determinant and $\mathrm{Pic}^0(X) = 0$ identify the third. Thus $G = E_0$. A contrary sequence outside the fixed open chart is impossible.

The same reasoning for the w-problem needs only one normalized line inclusion; its determinant identifies the other line. For a compact set of H-stable w-bundles, local universal families and smooth HE metrics give the analogous uniform statement at $L_1 \oplus F$.

On the finite side, every point over the affine vertex has zero cycle invariants. Its directed support is acyclic. Summing the real moments over successive source or sink subsets bounds each arrow norm squared by a fixed multiple of $|\zeta|$. There are finitely many supports. Consequently the entire finite fibre has coefficient radius $O(\sqrt{|\zeta|})$ and lies in the same small chart. Both actual and finite families classify the same central graded objects. The two coverage statements therefore identify the whole fibres on both sides.

For later use this is also uniform when the moving factor tends to its split class. If $F_n \to L_0 \oplus L_2$ at the central polarization, the fixed-polarization surface comparison [GSTW21, Theorem 4.11 and Corollary 4.15], together with zero charge, gives that gauge limit. For chamber extensions involving L_1 and F_n, normalize both extension maps. The limiting map involving L_1 splits off that line, and the map involving $L_0 \oplus L_2$ splits off at least one further line. The determinant identifies the remaining summand. The resulting limit is again E_0. Thus one fixed deep chart covers the moving fibres as their bases approach the deeper point; no uniform elliptic gap for separate two-factor backgrounds is required there.

4.5. **The singular K3 surface and common target.** Let S be a generic compact w-moduli K3 surface. Its local universal families define compatible Greb–Toma classifying maps to the central slope space. Scalar twists do not affect these maps, so they descend to S. The only nontrivial fibre is the projective line of extensions of L_0, L_2. By whole-fibre coverage it is the zero section of the two-arrow U(1) quotient. Normalize the reduced image to obtain T. The saturated-neighborhood and direct-image argument used for (21) identifies the local map with $T^*\mathbb{P}^1 \to \mathbb{C}^2/\{\pm 1\}$. Thus T is a normal projective K3 surface with one A_1 point, $S \to T$ is its minimal resolution, and $S^\circ = S \setminus C$ is the nonempty moduli of H-stable w-bundles. Here C is the exceptional (-2)-curve.

Every nearby numerical wall through H for the rank-three problem arises from a proper sum in Lemma 4.1. Its equation is therefore one of $\zeta_i = 0$. Walls not passing through H are avoided after shrinking the parameter ball. Each displayed wall is actual: the double-edge quotient realizes $\zeta_1 = 0$, and a spherical single-edge pair at nonzero internal level realizes $\zeta_0 = 0$ or $\zeta_2 = 0$. There are exactly six chambers.

Each sector contains integral ample classes $mH + pD_1 + qD_2$ with p not divisible by three. Their degree $-10p - 12q$ is coprime to three, so their full compact Gieseker moduli are the slope/HYM chamber moduli M_σ. Their universal families give central slope classifying maps. Their common image consists of the stable open set U, the classes $L_1 \oplus F$ for $F \in S^\circ$, and E_0. Every listed class occurs by the full local quotients. The point (27), dilated to the origin, proves that U is nonempty and dense. Its connectedness follows from irreducibility of the chamber space.

Fibres over U are points, those over S° are projective lines, and the deepest fibre is the connected projective fibre of $\mathcal{Q}_\zeta \to \mathcal{Q}_0$. Normalizing the common reduced image yields a normal projective fourfold Z. Connectedness and Stein factorization make this normalization bijective on points. Whole-fibre coverage precedes the use of properness to shrink to saturated neighborhoods. On those neighborhoods, the fixed holomorphic classifying families commute with central S-equivalence. Descending coordinate functions by $\pi_*\mathcal{O} = \mathcal{O}$ identifies the actual and finite target germs in both directions. The local model is (26) at E_0 and a smooth surface times the A_1 cone along S°. The quotient symplectic forms agree with the moduli forms, giving symplectic resolutions $M_\sigma \to Z$.

The assignment $F \mapsto L_1 \oplus F$ identifies T with the reduced singular locus of Z. Construct the map first on S using local universal families; it is constant on C and descends to the normal T. Along S° the product chart makes it a closed immersion. At E_0, the fixed space of the vertex-1 subtorus is the double-edge representation. Both the harmonic recursion and the equivariant Darboux map preserve this fixed space. Restriction of invariant rings onto its ring is surjective, since every double-edge invariant already has weight zero for the full torus. Hence the map is a closed immersion at the remaining point as well. Its image is reduced because T is normal, and the stratum classification identifies its support with the singular set. This proves the assertion about the reduced singular locus.

This constructs the full compact diagram. The two stable factors over S° have cross-Ext dimension $(a_1, w) = 2$, while the three lines at o have the full eight-dimensional doubled representation. The change of local model is exactly the change of singular stratum. The explicit fibres and their specialization are computed next.

5. Exceptional fibres and specialization

The six chambers have two types of deepest fibre. We first describe the two-plane fibre, because it shows directly how a moving exceptional curve acquires two components. We then calculate the remaining chambers. These finite configurations occur in Bielawski–Dancer [BD00, Example 6.10]. Section 4 identifies them with the *whole* fibres in our compact bundle moduli spaces.

5.1. **Two planes and a reduced degeneration of curves.** Take $\zeta_0 = a > 0$, $\zeta_2 = b > 0$, and $\zeta_1 = -a - b$. Over the affine vertex every oriented-cycle invariant vanishes. In particular,

$$zz' = ww' = 0, \qquad u_i\varphi_j = 0 \quad (1 \le i, j \le 2).$$

Thus $u = 0$ or $\varphi = 0$, and a nonzero triangle of arrows must be acyclic. If $\varphi \neq 0$, the positive moment at vertex 2 forces $w \neq 0$; the cycle equations then force $z' = w' = 0$. This

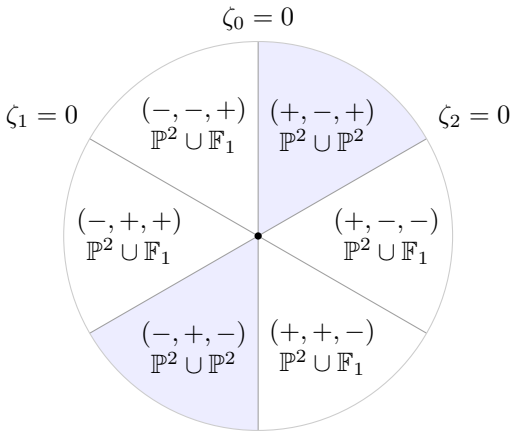


FIGURE 1. The six sectors in the moment parameter plane. Signs are ordered $(\zeta_0, \zeta_1, \zeta_2)$. The shaded sectors have two planes meeting at a point; in the other sectors the two surfaces meet along a projective line.

component, including its closure, is

$$u = z' = w' = 0, \qquad |z|^2 + |\varphi|^2 = a, \qquad |w|^2 = b + |\varphi|^2. \tag{28}$$

Eliminate the nonzero coordinate w by complex gauge. The remaining quotient is $[z : \varphi] \in \mathbb{P}^2$. Call it P_0: its objects have L_0 as subobject and the spherical extension of L_1, L_2 as quotient. The other component P_2 has $\varphi = z' = w' = 0$, $z \neq 0$, and coordinates $[w : u] \in \mathbb{P}^2$. The components meet at the single point p with $u = \varphi = 0$ and $z, w \neq 0$.

These are all the components, since they exhaust the acyclic supports. Near p fix z, w by complex gauge and solve the two complex equations for z', w'. The remaining smooth coordinates are (u, φ). Thus $P_0 = \{u = 0\}$ and $P_2 = \{\varphi = 0\}$ meet transversely. The moving exceptional divisor D is locally $\varphi(u) = 0$, with contraction to the A_1 surface given by

$$A = u \otimes \varphi, \qquad \operatorname{tr} A = 0.$$

In particular, D has an ordinary double point at p. It is irreducible: its projective-line family over the connected open surface S° is dense, and a fibre over o has dimension two, not three.

Fix nonzero $b_0 \in \mathbb{C}^2$ and $\beta_0 \in (\mathbb{C}^2)^*$ with $\beta_0(b_0) = 0$. Over $A = tb_0 \otimes \beta_0$, write $u = \lambda b_0$, $\varphi = \nu\beta_0$, $\lambda\nu = t$, and $z' = w' = 0$. The compact quotient has coordinates

$$[Z_0 : Z_1] = [z : \nu w], \qquad [W_0 : W_1] = [w : \lambda z], \qquad Z_1 W_1 = t Z_0 W_0 \tag{29}$$

in $\mathbb{P}^1 \times \mathbb{P}^1$. Stability makes each homogeneous pair nonzero. On their four affine charts the displayed expressions and the equation recover the original quotient coordinates, including the endpoints where z or w vanishes.

Equation (29) is a relative effective Cartier divisor of fixed bidegree $(1, 1)$, containing no entire ambient fibre. It is therefore proper and flat over a disk. For $t \neq 0$ its fibre is $\mathbb{P}^1$; for $t = 0$ it is the reduced union of a line in P_0 and a line in P_2. At their crossing the equation is $xy = t$. Whole-fibre coverage from Section 4 embeds this compact family into the actual moduli space after shrinking the disk. Consequently the specialization maps in integral homology and fibre cohomology are

$$H_2(\mathbb{P}^1, \mathbb{Z}) \longrightarrow H_2(P_0 \cup P_2, \mathbb{Z}), \quad 1 \longmapsto (1, 1), \qquad (n_0, n_2) \longmapsto n_0 + n_2. \tag{30}$$

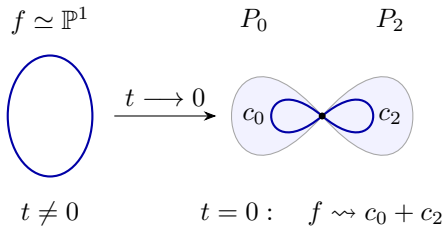


FIGURE 2. Incidence schematic for the proper family (29). The marked curves in the two planes each occur with multiplicity one; the local equation at their common point is $xy = t$.

This is a multiplicity-one geometric specialization, not merely an identity of rational divisor pairings.

5.2. **The other four chambers.** Two symmetries reduce the calculation to one more sector. Interchanging every arrow with its paired reverse and inverting the vertex bases identifies the model at ζ with that at $-\zeta$. The signed cotangent lift

$$(z, z', w, w', u, \varphi) \longmapsto (w, w', z, z', \varphi, -u)$$

interchanges vertices 0 and 2. Both operations preserve the complex zero equations and the flat metric. They are symmetries of the finite models; they need not arise from automorphisms of the fixed K3 surface.

Consider $\zeta_0 = a > 0$, $\zeta_1 = b > 0$, $\zeta_2 = -a - b$. A maximal acyclic support must use the double edge in direction $2 \to 0$. The other edges have one of the orientations $2 \to 1 \to 0$ or $2 \to 0 \to 1$. In the first case the coordinates are (φ, w', z) and

$$|\varphi|^2 + |z|^2 = a, \qquad |w'|^2 - |z|^2 = b.$$

Eliminating the nonzero w' leaves $\mathbb{P}^2$. In the second case they are (φ, z', w') and

$$|\varphi|^2 - |z'|^2 = a, \qquad |z'|^2 + |w'|^2 = b.$$

Here $\varphi \neq 0$ and $(z', w') \neq 0$. The effective two-torus has weights $(1, 0), (-1, 1), (0, 1)$ on these three groups of coordinates. The quotient is the first Hirzebruch surface $\mathbb{F}_1$: over $[\varphi] \in \mathbb{P}^1$ the coordinate z'/w' transforms as a coordinate on $\mathcal{O}_{\mathbb{P}^1}(-1)$, and its compactification adds the infinity section.

The two surfaces meet at $z = z' = 0$, where $\varphi, w' \neq 0$. Their intersection is a line in $\mathbb{P}^2$ and the negative section in $\mathbb{F}_1$. For an explicit local check, fix $\varphi = (1, q)$ and $w' = 1$. The complex equations give $u_1 = zz' - qu_2$ and $w = -zz'$. In the smooth coordinates (z, z', u_2, q) the surfaces are $\{z' = u_2 = 0\}$ and $\{z = u_2 = 0\}$. Their intersection is clean, with tangent space exactly the q-line; the second φ chart gives the same description at infinity.

We have proved the complete table

Signs of $(\zeta_0, \zeta_1, \zeta_2)$	Components	Intersection
$(+,-,+),\ (-,+,-)$	$\mathbb{P}^2,\ \mathbb{P}^2$	transverse point
$(+,+,-),\ (-,-,+)$	$\mathbb{P}^2,\ \mathbb{F}_1$	line / negative section
$(-,+,+),\ (+,-,-)$	$\mathbb{P}^2,\ \mathbb{F}_1$	line / negative section

Each component is Lagrangian: its coordinate subspace contains at most one direction from each symplectically paired set of arrows, and has half the dimension after reduction.

5.3. **Intersections and adjacent walls.** Write P, R for the plane and ruled surface in the second configuration. Their normal bundles are T^*P, T^*R, giving self-intersections 3 and 4. For their clean intersection $C \simeq \mathbb{P}^1$ the excess bundle is $T_M/(T_P + T_R)|_C \simeq T^*C$, by the symplectic form. The excess intersection formula gives

$$P \cdot R = -2, \qquad B' = \begin{pmatrix} 3 & -2 \\ -2 & 4 \end{pmatrix}. \tag{31}$$

The negative sign comes from a curve of excess intersection, rather than an isolated transverse complex intersection.

Mayer–Vietoris for $P \cup_C R$ gives $H_0 = \mathbb{Z}$ and $H_2 = H_4 = \mathbb{Z}^2$, with all other reduced homology zero. The map from $H_2(C)$ is primitive because its first component is a line. A regular neighborhood W retracts to this union. Poincaré–Lefschetz duality identifies $H^4(W, \partial W) \to H^4(W)$ with B'. Its Smith invariants are $1, 8$. Thus the boundary groups are those in (39) below, with quotient functional $(n_P, n_R) \mapsto 6n_P + n_R \pmod 8$. This is a calculation of the resolved-neighborhood boundary. The global primitive embeddings used in the next section are calculated in the two-plane chamber.

Away from the origin, the walls $\zeta_0 = 0$ and $\zeta_2 = 0$ have the doubled three-arrow U(1) normal model. They contract an isolated plane. The wall $\zeta_1 = 0$ has the doubled two-arrow model and contracts the moving projective-line family. Its normal singularity is A_1. The factor classification and whole-fibre coverage show that this list includes every exceptional component in the chosen parameter neighborhood.

6. Integral cycles and their attachment

All cohomology in this section has integral coefficients unless otherwise specified. The Mukai marking identifies the exceptional curve classes, but rational intersection formulas alone do not determine how their neighborhoods attach to the complement. We calculate both the primitive integral classes and the resulting boundary homomorphisms.

6.1. **Primitive classes in the isolated example.** On P_+ the universal extension has K-class $b\,\mathcal{O}(1) + a$, up to tensoring by a base line bundle. For $x \in v^\perp$ this ambiguity disappears from (14); hence

$$\theta_+(x)|_{P_+} = (a, x)h, \qquad \theta_-(x)|_{P_-} = -(a, x)h.$$

We identify a rational vector in $v^\perp$ with its curve functional by the Mukai pairing. A plane line therefore has class

$$\ell_\pm = \pm(a - v/2), \qquad q(\ell_\pm) = -5/2, \qquad \rho_\pm = \theta_\pm\big(\pm(2a - v)\big). \tag{32}$$

The integral vector $w_0 = (9, 3D, -10)$ lies in $v^\perp$ and satisfies $(a, w_0) = 1$. It proves that each line is primitive. The vector $2a - v = (-1, -D, 6)$ is primitive; its pairings with $v^\perp$ are even, and its pairing with w_0 is two. Thus $q(\rho_\pm) = -10$ and $\rho_\pm$ has divisibility two. The restriction of $\theta_\pm(w_0)^2$ to the plane is h^2, which also proves primitivity of its surface class.

For a K3$^{[2]}$-type fourfold the polarized Fujiki identities are

$$\begin{aligned} \int_M \alpha\beta\gamma\delta &= q(\alpha,\beta)q(\gamma,\delta) + q(\alpha,\gamma)q(\beta,\delta) + q(\alpha,\delta)q(\beta,\gamma), \\ \int_M c_2(M)\alpha\beta &= 30q(\alpha,\beta). \end{aligned} \tag{33}$$

The rational map $\operatorname{Sym}^2 H^2(M,\mathbb{Q}) \to H^4(M,\mathbb{Q})$ is an isomorphism. Since $q(\ell) = -5/2$, the class $\ell^2/2 + c_2(M)/24$ pairs with $\alpha\beta$ as $q(\ell,\alpha)q(\ell,\beta)$, exactly the restriction pairing on the plane. Nondegeneracy gives the classical formula

$$[P_\pm] = \frac{\ell_\pm^2}{2} + \frac{c_2(M_\pm)}{24} = \frac{\rho_\pm^2}{8} + \frac{c_2(M_\pm)}{24}. \tag{34}$$

See [HT09], with the attribution there to O'Grady. The cotangent normal bundle gives $[P]^2 = 3$; from $T_M|_P = T_P + T_P^*$ in K-theory one also gets $\int_P c_2(M) = -3$. Formula (34) is rational. The integral primitivity assertion follows from w_0, without assuming an integral symmetric-square presentation.

Totaro's torsion-freeness theorem [Tot20, Theorem 3.1] and the coefficient of degree two in Göttsche's Hilbert-scheme formula [Göt90] give free even cohomology of ranks $1, 23, 276, 23, 1$ for $M_\pm$. The contraction identifies Z homeomorphically with $M_\pm/P_\pm$: the map from this compact quotient to the Hausdorff target is bijective. The plane inclusion is a cofibration, and its restrictions in degrees 0, 2 and 4 are surjective. The pair sequence therefore gives

Space	b_0	b_2	b_4	b_6	b_8
$M_\pm$	1	23	276	23	1
Z	1	22	275	23	1
$M_\pm \setminus P_\pm$	1	23	275	22	0

All other groups vanish, and these groups are torsion-free. The pullback from Z is an injective ring map whose image is

$$\mathbb{Z}\cdot 1 \;\oplus\; \left(\bigoplus_{k>0} \ker\bigl(H^k(M_\pm) \longrightarrow H^k(P_\pm)\bigr) \right). \tag{35}$$

Its multiplication is inherited from the chamber ring; the local singularity alone does not provide that ring as additional output.

6.2. **The two interacting curve and surface classes.** Return to the chamber $\zeta_0, \zeta_2 > 0$, and put $E = P_0 \cup P_2$. Let c_i be an effective line in P_i and f a moving projective-line fibre. The universal extensions on these fibres give

$$\begin{gathered} l_i = a_i - v/2 \quad (i = 0, 2), \qquad c_0 = -l_0, \quad c_2 = -l_2, \quad f = a_1, \\ f = c_0 + c_2, \qquad q(c_0) = q(c_2) = -5/2, \quad q(c_0, c_2) = 3/2, \quad q(f) = -2. \end{gathered} \tag{36}$$

Indeed the restriction to P_i is $\theta_v(x)|_{P_i} = -(a_i, x)h_i$. Over a moving fibre the extension is $F\,\mathcal{O}(1) + L_1$, so its restriction is $-(w,x)h = (a_1,x)h$. This calculation uses a fixed F and needs no global ordinary universal family on S°. Equation (30) realizes the class relation by a reduced family of actual curves.

Use the primitive lattice embedding (22). The integral vectors

$$x = (0, f_2 - f_3, 0), \qquad y = (0, -3f_2, -1)$$

are in $v^\perp$, with pairings $(0, 1, -1)$ and $(1, -2, 1)$ against (a_0, a_1, a_2). The same witnesses exist for every primitive embedding of $\mathrm{NS}(X)$: unimodularity of $H^2(X,\mathbb{Z})$ makes the restriction $H^2(X,\mathbb{Z}) \to \mathrm{NS}(X)^*$ surjective, so the prescribed pairings of their degree-two components can always be realized. For $\alpha = \theta_v(x)$ and $\beta = \theta_v(y)$,

$$\alpha|_E = (0, h_2), \qquad \beta|_E = (-h_0, -h_2), \qquad \alpha^2|_E = (0, h_2^2), \quad \beta^2|_E = (h_0^2, h_2^2). \tag{37}$$

Both restriction matrices are unimodular. Thus the two line classes span a primitive subgroup of $H_2(M)$, and the two plane classes span a primitive subgroup of $H_4(M)$.

The cotangent normal bundles and the transverse intersection at p give

$$B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}. \tag{38}$$

Equivalently, $[P_i] = l_i^2/2 + c_2(M)/24$ by the argument for (34); the off-diagonal intersection is $q(l_0, l_2)^2/2 - 1/8 = 1$. Here the geometric intersection and the rational Fujiki calculation agree, while (37) supplies the integral information.

6.3. **The neighborhood boundaries.** For one plane the boundary is the sphere bundle $L = S(T^*\mathbb{P}^2)$. Its Euler class is $3h^2$. The Gysin sequence gives $H^j(L) = \mathbb{Z}$ for $j = 0, 2, 5, 7$, $H^4(L) = \mathbb{Z}/3$, and zero otherwise. In particular $H^3(L) = 0$, although $H_3(L) = \mathbb{Z}/3$.

For two transverse planes take a closed regular neighborhood W with smoothed boundary L. It is the plumbing of their normal disk bundles and retracts to $E \simeq \mathbb{P}^2 \vee \mathbb{P}^2$. Poincaré–Lefschetz duality makes the middle map $H^4(W, L) \to H^4(W)$ the matrix B. The pair sequence, together with $\det B = 8$ and the gcd one of its entries, gives

$$H^j(L) = \begin{cases} \mathbb{Z}, & j = 0, 7, \\ \mathbb{Z}^2, & j = 2, 5, \\ \mathbb{Z}/8, & j = 4, \\ 0, & \text{otherwise.} \end{cases} \tag{39}$$

The quotient map on $H^4(W) = \mathbb{Z}^2$ is

$$(n_0, n_2) \longmapsto n_0 - 3n_2 \pmod 8. \tag{40}$$

Its kernel is $B\mathbb{Z}^2$: it kills both columns and is onto a group of order $\det B$. This L is a smooth boundary in the resolution. The target has a positive-dimensional singular stratum through o, so L is not the link of an isolated target singularity.

Alexander–Lefschetz duality gives $H^k(M, M \setminus E) = H_{8-k}(E)$. The primitive line, plane and point inclusions just proved show that $V = M \setminus E$ has torsion-free even cohomology, with ranks $1, 23, 274, 21$ in degrees $0, 2, 4, 6$. In particular,

$$H^4(V) = H^4(M)/(\mathbb{Z}[P_0] + \mathbb{Z}[P_2]). \tag{41}$$

Notice that V retains the moving exceptional divisor away from E.

6.4. **Calculated attachments on integral generators.** We now specify the map from the complement to the boundary. The input includes the complement ring with its distinguished universal-index classes. The following formulas evaluate the attachment without asking for an ambient lift of a given class.

For $z \in K^0_{\mathrm{top}}(X)$ define the family index

$$I_z = p_{M!}\big(p_X^*(z^\vee) \otimes \mathcal{E}\big) \in K^0_{\mathrm{top}}(M), \qquad \chi(F, z) = -(v(F), v(z)). \tag{42}$$

Here $p_!$ is the complex K-theory pushforward. Markman's integral-generation theorem [Mar07, Theorem 1], the integral Künneth decomposition and the perfect Euler pairing on $K^0_{\mathrm{top}}(X)$ show that the Chern classes of these indices generate $H^*(M, \mathbb{Z})$. Thus degree four is generated by the classes $c_2(I_z)$ and $c_1(I_z)c_1(I_{z'})$. Rational Chern characters would not suffice for this conclusion.

One normalization removes the base-line ambiguity of the universal family on the exceptional set. For the isolated example, let F, Q be its subobject and quotient in the chosen chamber. Their Mukai vectors span a primitive rank-two sublattice: the rank-one factor and

the primitive D-coordinate give a unit minor. Unimodularity supplies a topological class y with $\chi(F,y)=0$, $\chi(Q,y)=1$. On the plane

$$\mathcal{E}=(F\,\mathcal{O}(1)+Q)\otimes L,\qquad I_y|_P=L.$$

The global index I_y has rank one. Tensor $\mathcal{E}$ by $\det(I_y)^{-1}$ to remove L. For this normalized topological universal class,

$$c(I_z)|_P=(1+h)^{m_z},\qquad m_z=\chi(F,z). \tag{43}$$

The powers are interpreted as total Chern classes of virtual bundles and allow negative integers.

For the interacting planes the class $y=(0,-f_2,0)$ has Euler pairings $(0,1,0)$ with (a_0,a_1,a_2). The same global determinant normalization removes the line twists on *both* planes, including their common point. It gives

$$c(I_z)|_{P_i}=(1+h_i)^{m_{i,z}},\qquad m_{i,z}=\chi(L_i,z),\quad i=0,2. \tag{44}$$

This normalization is topological; it makes no claim of an algebraic lift for the auxiliary Brauer class considered in the next section.

Proposition 6.1. *Let $A=H^*(V,\mathbb{Z})$ be the complement ring with the restrictions of the normalized index Chern classes just specified.*

For an isolated plane, the actual restriction $\kappa: A^4\to H^4(L)=\mathbb{Z}/3$ is determined by

$$\begin{aligned}\kappa(c_2(I_z)|_V)&=\binom{m_z}{2},\\ \kappa(c_1(I_z)c_1(I_{z'})|_V)&=m_zm_{z'}\end{aligned}\qquad\pmod 3, \tag{45}$$

and restriction identifies

$$H^4(M)=\{(u,N)\in A^4\oplus\mathbb{Z}: N\equiv\kappa(u)\pmod 3\}. \tag{46}$$

For the two-plane chamber, the actual restriction $\kappa: A^4\to H^4(L)=\mathbb{Z}/8$ is determined by

$$\begin{aligned}\kappa(c_2(I_z)|_V)&=\binom{m_{0,z}}{2}-3\binom{m_{2,z}}{2},\\ \kappa(c_1(I_z)c_1(I_{z'})|_V)&=m_{0,z}m_{0,z'}-3m_{2,z}m_{2,z'}\end{aligned}\qquad\pmod 8, \tag{47}$$

and restriction identifies

$$H^4(M)=\{(u,n_0,n_2)\in A^4\oplus\mathbb{Z}^2: n_0-3n_2\equiv\kappa(u)\pmod 8\}. \tag{48}$$

Both attachment maps are surjective.

Proof. The relative group $H^5(M,V)=H_3(E)$ vanishes in both cases. Thus $H^4(M)\to A^4$ is onto, and the restricted degree-four index monomials generate A^4. Equations (43) and (44) give their integer evaluations on the plane or planes. Passing to the boundary gives exactly (45) and (47).

To check every relation among these generators, let a degree-four integral polynomial in the restricted Chern classes vanish on V. Its ambient polynomial class lies in the relative image $H^4(M,V)$, generated by the plane classes. Its evaluations on E therefore lie in $3\mathbb{Z}$ or $B\mathbb{Z}^2$, respectively. The prescribed boundary functional kills these subgroups. Consequently the formulas are independent of the chosen polynomial expression in A^4 and define its actual boundary restriction.

Apply Mayer–Vietoris to V and an open regular neighborhood of E. Their overlap retracts to L, and $H^3(L)=0$. The restriction from M is therefore injective in degree four, with image

the kernel of the difference of the two boundary maps. This is precisely the displayed fibre product. Finally $\theta(w_0)^2$ evaluates to one in the isolated case, while α^2 evaluates to $(0,1)$ in the interacting case. Their images are units modulo three and eight. □

The construction retains the complement ring and its marked index classes as input. It recovers the integral degree-four attachment, including the torsion boundary information that is absent from Betti numbers. A change of index basis or universal normalization changes the formulas for the marked generators, but not the restriction map they calculate.

7. The projective-line family over the open stratum

Let $S \to T$ be the auxiliary K3 resolution, with exceptional (-2)-curve C, and identify $S^\circ = S \setminus C = T \setminus \{o\}$. In the two-plane chamber write $D^\circ = D \setminus E$. The map $\pi : D^\circ \to S^\circ$ is a projective-line bundle. It has trivial monodromy on fibre cohomology, yet a nonzero algebraic Brauer obstruction. This distinction is visible in the global topology.

7.1. **Algebraic, analytic and topological lifts.** For a stable bundle F of vector w, the groups $\mathrm{Hom}(L_1, F)$ and $\mathrm{Ext}^2(L_1, F)$ vanish, while $\dim \mathrm{Ext}^1(L_1, F) = 2$. Local universal w-families therefore give rank-two extension bundles. Their projectivizations descend to D°, with the order-two Brauer class of the universal w-family. This class remains nonzero on S°: for a regular integral scheme the Brauer group injects into that of its function field [Gro66, Corollaries 1.8 and 1.10]. Consequently D° has no presentation as the projectivization of an ordinary *algebraic* rank-two vector bundle on S°.

The curve C is primitive in the unimodular K3 lattice, since its square is -2. The Gysin sequence gives

$$H^1(S^\circ) = H^3(S^\circ) = H^4(S^\circ) = 0, \qquad H^2(S^\circ) = H^2(S)/\mathbb{Z}[C] \simeq \mathbb{Z}^{21}. \tag{49}$$

The connected group $\mathrm{PGL}_2(\mathbb{C})$ acts trivially on $H^0(\mathbb{P}^1)$ and $H^2(\mathbb{P}^1)$. The fibre cohomology local systems are thus constant. There is also no topological obstruction in degree three to lifting the projective bundle. The class $u = \alpha|_{D^\circ}$ supplies an explicit integral hyperplane class, because $\alpha \cdot f = 1$.

An ordinary holomorphic lift exists as well. Siu's top-degree coherent-cohomology theorem for a connected noncompact complex manifold [Siu69] gives $H^2(S^\circ, \mathcal{O}) = 0$. Together with $H^3(S^\circ, \mathbb{Z}) = 0$, the exponential sequence gives $H^2(S^\circ, \mathcal{O}^*) = 0$. The obstruction from $1 \to \mathcal{O}^* \to \mathrm{GL}_2(\mathcal{O}) \to \mathrm{PGL}_2(\mathcal{O}) \to 1$ therefore vanishes. There is no contradiction: on this nonproper surface analytification need not retain an algebraic Brauer obstruction.

7.2. **The divisor class, including its multiplicity.** We need the precise class of D to calculate the ring of D°. Put $A = L_1$. The inequality $\zeta_1 < 0$ means $\mu_\eta(A) > \mu_\eta(E)$, so $\mathrm{Hom}(A, E) = 0$ throughout the chamber. Moreover $\chi(A, E) = -(a_1, v) = 0$.

A nonzero map $E \to A$ has image $A(-B) \otimes I_Z$ for an effective divisor B and a zero-dimensional scheme Z. At H its image is a torsion-free quotient of a semistable slope-zero bundle, so has nonnegative slope. As a subsheaf of A it has slope $-H \cdot B$. Ampleness forces $B = 0$. The kernel has slope zero and is semistable. The complete factor and defect classification of Lemma 4.1 then forces $Z = \varnothing$ and makes the kernel a bundle of vector w. Conversely the extension fibres, including both deepest planes, have a quotient to A. Hence $\mathrm{Hom}(E, A) \neq 0$ precisely on D.

The perfect complex $Rp_* R\,\mathrm{Hom}(A, \mathcal{E})$ has amplitude $[1, 2]$, virtual rank zero, and is acyclic off D. Locally write it as $K^1 \xrightarrow{d} K^2$. Its determinant section lies in $\det(K^1)^{-1} \otimes \det(K^2)$, whose first Chern class is $\theta_v(a_1)$ by Grothendieck–Riemann–Roch and (14).

We check that the section vanishes simply at a generic point of D. There the bundle fits into $0 \to F \xrightarrow{j} E \xrightarrow{q} A \to 0$, with F H-stable. The extension sequence, $\operatorname{Hom}(F,A) = \operatorname{Hom}(A,F) = 0$, and simplicity give

$$\operatorname{Hom}(F,E) = \mathbb{C}j, \quad \operatorname{Hom}(E,A) = \mathbb{C}q, \quad \operatorname{Hom}(E,E) = \mathbb{C},$$
$$\operatorname{Ext}^1(E,A) = \mathbb{C}, \quad \operatorname{Ext}^2(E,A) = 0.$$

In the exact sequence

$$\operatorname{Ext}^1(E,E) \xrightarrow{q_*} \operatorname{Ext}^1(E,A) \longrightarrow \operatorname{Ext}^2(E,F) \xrightarrow{j_*} \operatorname{Ext}^2(E,E),$$

the last arrow is an isomorphism: its Serre dual sends id_E to j between one-dimensional spaces. Thus q_* is onto. For an infinitesimal deformation δ of E, $q_*\delta$ is the obstruction to extending q. Serre duality identifies this obstruction with the first variation of the local differential from $\operatorname{Ext}^1(A,E)$ to $\operatorname{Ext}^2(A,E)$, up to sign. After cancelling acyclic summands, d is a scalar function with nonzero derivative. Its zero is simple. These tangents preserve the determinant because $H^1(X,\mathcal{O}_X) = 0$.

Since D is irreducible and is the whole divisorial support, we have proved

$$[D] = \theta_v(a_1), \qquad D|_{P_i} = -h_i, \qquad D \cdot f = -2. \tag{50}$$

The multiplicity argument is needed: the last two intersection numbers by themselves would not identify the global divisor class.

7.3. **The ring over the open stratum.** Use the integral isotropic Mukai map $\theta_w : w^\perp/\mathbb{Z}w \to H^2(S,\mathbb{Z})$ induced by the same w-family as in the construction. With x from (37), set

$$z = a_1 + 2x, \qquad d = \theta_w(z)|_{S^\circ}.$$

Direct pairing gives $z \in v^\perp$, $(z,a_1) = 0$ and $(z,w) = 0$. The universal extension over D° gives

$$\theta_v(z)|_{D^\circ} = \pi^*\theta_w(z)|_{S^\circ}. \tag{51}$$

Here is the descent argument for this global identity. On the universal gerbe, use the normalized rational Chern character of the twisted family; equivalently divide the Chern character of a quasi-universal family by its similitude. Additivity in the universal extension gives (51) in rational cohomology. A tautological line, a base line or a rational choice for the torsion gerbe affects the degree-two pushforward only through (z,a_1), (z,w) or (z,v), all zero. Thus the identity descends globally, rather than only on contractible charts. Both sides are integral. Leray–Hirsch with u and (49) shows that $H^2(D^\circ,\mathbb{Z})$ is torsion-free, so the rational equality is integral.

Combining it with (50) gives

$$D|_{D^\circ} = -2u + \pi^*d. \tag{52}$$

By adjunction this is the relative canonical class: M and S° have trivial canonical bundles. Choose a topological rank-two lift $\mathcal{W}$ of the projective bundle, in the quotient convention for $\mathbb{P}(\mathcal{W})$. Its hyperplane class differs from u by a pullback from the base. Tensoring $\mathcal{W}$ with a topological line bundle removes this difference. The normalized lift has relative canonical class $-2u + \pi^*c_1(\mathcal{W})$, so injectivity of π^* gives $c_1(\mathcal{W}) = d$. Also $c_2(\mathcal{W}) = 0$ by (49). The integral projective-bundle formula now yields

$$H^*(D^\circ,\mathbb{Z}) = H^*(S^\circ,\mathbb{Z})[u]/(u^2 - du). \tag{53}$$

This calculation specifies the hyperplane normalization. It is consistent with the nonzero algebraic Brauer class and the trivial fibre-cohomology monodromy: these record different aspects of the same family.

8. Harmonic collapse and complete rescaled limits

The classes calculated above determine exact periods. To determine diameters, one must also compare paths in the full compact moduli space. We first construct compatible local comparisons and explain why short ambient paths remain in them. This permits a relative estimate even when one exceptional plane is much smaller than the other.

8.1. **Exact periods of the reconstructed cycles.** Write

$$b_\eta = \left(0, [\omega_\eta], \frac{c_1(v)\cdot[\omega_\eta]}{\operatorname{rank} v}\right).$$

Equation (15) gives $[\Omega_\eta] = 2\pi^2\theta_v(b_\eta)$. In the interacting family $(a_i, b_\eta) = \deg_\eta L_i - \mu_\eta(v) = -\zeta_i/(2\pi)$. Pairing with (36) proves

$$\begin{gathered}\operatorname{Area}(c_0) = \pi a, \qquad \operatorname{Area}(c_2) = \pi b, \qquad \operatorname{Area}(f) = \pi(a+b),\\ \operatorname{Vol}(P_0) = \frac{(\pi a)^2}{2}, \qquad \operatorname{Vol}(P_2) = \frac{(\pi b)^2}{2}.\end{gathered} \tag{54}$$

The volume formula is $\int_P \Omega_\eta^2/2$. These are exact identities for the actual harmonic metric; they do not require its induced plane metric to be Fubini–Study. The reduced specialization of curves is reflected in exact additivity of their areas.

For the isolated example put $\kappa_t = 44\pi|t|/3$. Its signed line class (32) similarly gives

$$\operatorname{Area}(\ell_\pm) = \pi\kappa_t, \qquad \operatorname{Vol}(P_\pm) = \frac{(\pi\kappa_t)^2}{2}. \tag{55}$$

The same κ_t is the absolute moment level in (20).

8.2. **Compatible comparisons over the entire exceptional set.** At the deeper point use the fixed background $E_0 = L_0 \oplus L_1 \oplus L_2$. Its chart Φ_η has the full model (26) and the estimate (13), with radius $r = \|x\|$. The common radius and constant are independent of ratios of positive eigenvalues of the orbit operator $C_x^* C_x$. Section 4 proves that the chart covers whole fibres and commutes with the contraction to Z.

For F in a compact subset of S°, local universal families give smoothly varying central HE connections on $L_1 \oplus F$. The elliptic complement has a uniformly positive lower bound on that compact set. Harmonic projections and the finite-regularity estimates vary smoothly, so finitely many charts have uniform constants. Their full harmonic representations are

$$U \oplus \left(\mathbb{C}^2 \oplus (\mathbb{C}^2)^*\right), \qquad \dim_{\mathbb{C}} U = 2, \qquad \kappa = 2\pi(2\mu_\eta(v) - \deg_\eta F) = -\zeta_1. \tag{56}$$

The effective group is a circle, trivial on U. The factor two in the target comes from the rank of F; the multiplicity-space moment still uses ordinary trace.

Near o these two-factor gaps need not remain positive. Instead choose one fixed smaller neighborhood of o covered by the three-factor chart. Its coverage includes moving fibres whose base tends to o, by the normalized-map compactness argument of Section 4. The remaining part of the contracted locus is compact in S° and admits the preceding finite cover.

On an overlap the transition is exactly $h_{ij,\eta} = \Phi_{j,\eta}^{-1}\Phi_{i,\eta}$. Both charts classify the same holomorphic bundles; HE and finite balancing change their presentations, not their isomorphism classes. The transitions are holomorphic on the complex quotient charts and commute with

the contraction. A base-line twist of a local universal family does not change either assertion. For $\varepsilon_i = C_i(|\eta| + r_i^2) < 1/2$, comparison with the same actual metric gives

$$\frac{1-\varepsilon_i}{1+\varepsilon_j} g_i \le h^*_{ij,\eta} g_j \le \frac{1+\varepsilon_i}{1-\varepsilon_j} g_i. \tag{57}$$

This relative statement remains valid when positive orbit eigenvalues become arbitrarily small. A bound for ordinary derivatives of balancing gauges is not required.

8.3. **From tensor comparison to ambient distances.** Equip each $\mathcal{Q}_\zeta$ with the length distance d_ζ obtained from continuous paths which are smooth in representation-type strata on a finite subdivision, with finite one-sided lengths at endpoints. The complex zero fibre has the module-simple point (27). The finite intrinsic metric theorem [Wue26a, Section 5] therefore applies: these are proper geodesic spaces with the compact quotient topology, their distances vary continuously with balanced representatives and levels, and every target point is approximated at nearby levels. The quotient norm radius is 1-Lipschitz.

Shrink the actual chart to have squared-metric bounds $1/2$ and 2 on $r < 2R$. Closed radial sublevels are compact in the quotient and their images are closed in the compact Hausdorff moduli space. An actual path starting at radius $r_0 < R$ and leaving the radius-R image must pay at least

$$\frac{R - r_0}{\sqrt{2}} \tag{58}$$

before its first exit. A segment of length L remaining in the chart has radius at most $r_0 + \sqrt{2}L$. This applies to paths in the entire moduli space and excludes a shortcut through a distant part of that space.

Here is the resulting complete rescaling statement, with its basepoint data made explicit. Let $\eta_n \to 0$ be generic, $\delta_n \downarrow 0$, and

$$\nu_n = \frac{\zeta(\eta_n)}{\delta_n^2} \longrightarrow \nu. \tag{59}$$

Let p_n be balanced representatives at levels ν_n converging, after unitary adjustment, to a representative p at level ν. Then

$$\big(M_{\sigma(\eta_n)}, \delta_n^{-1} d_{g_{\eta_n}}, \Phi_{\eta_n}([\delta_n p_n])\big) \longrightarrow (\mathcal{Q}_\nu, d_\nu, [p]) \tag{60}$$

in pointed Gromov–Hausdorff distance.

For completeness, homogeneity identifies the current finite model with $\delta_n \mathcal{Q}_{\nu_n}$. Finite near-minimizers on a bounded rescaled ball have bounded radius, by its 1-Lipschitz property, and scale into the actual chart. Conversely an actual competitor of length $O(\delta_n)$ cannot pay the fixed exit cost (58); its radius is $O(\delta_n)$. Applying (13) to both types of competitors and taking both infima gives relative distance error

$$O_L(|\eta_n| + \delta_n^2) \tag{61}$$

on each bounded rescaled ball. This comparison also holds for pairs much closer than δ_n. The inverse chart covers every actual bounded ball, and the model distance continuity identifies the full complete target. The estimate compares the actual metric with the *current* quotient $\mathcal{Q}_{\nu_n}$; it does not prescribe a rate for $\mathcal{Q}_{\nu_n} \to \mathcal{Q}_\nu$.

At a fixed point of S° the same proof uses (56). It gives the complete product of the flat harmonic tangent space U and the Eguchi–Hanson quotient $T^*\mathbb{P}^1$ at nonzero normalized level, or $U \times (\mathbb{C}^2/\{\pm 1\})$ at zero level. The statements are uniform on compact subsets of S°.

8.4. **Two scales in the deepest fibre.** Let $C_i(a,b)$ denote the finite planes identified with P_i, and put $\rho = a+b$. Their balanced radii and finite ambient diameters are $O(\sqrt{\rho})$, by the finite core bound [Wue26a, Section 6]. The preceding first-exit argument, now at scale $\sqrt{\rho}$, gives

$$\operatorname{diam}_{g_\eta} P_i = \sqrt{\rho}\,(1+O(\rho))D_i(a/\rho, b/\rho), \tag{62}$$

where $D_i(A,B)$ is the ambient diameter of the indicated plane in the complete quotient $\mathcal{Q}_{(A,-A-B,B)}$. The bounded inverse of the linear parameter map makes $|\eta| = O(\rho)$ in this chamber. The error is relative even when $D_i \to 0$: shorter competitors stay in the same radius-$O(\sqrt{\rho})$ region, so no division by a weak diameter occurs.

An explicit calculation separates the scales. On C_0, use (28) and fix w positive real. Before taking the remaining common phase quotient of (z,φ), minimizing over the eliminated phase gives

$$|dz|^2 + |d\varphi|^2 + \frac{(\operatorname{Re}\langle\varphi, d\varphi\rangle)^2}{b+|\varphi|^2} - \frac{(\operatorname{Im}\langle\varphi, d\varphi\rangle)^2}{b+2|\varphi|^2}. \tag{63}$$

It lies between one half and twice the flat form on (z,φ). Infimizing over the remaining phase preserves these inequalities. Thus the induced plane metric is bounded between one half and twice Fubini–Study at line area πa. Its intrinsic diameter is at most $\pi\sqrt{a}/\sqrt{2}$, an ambient upper bound as well.

For a lower bound use the points with $z=0$, $\varphi = \sqrt{a}(1,0)$ and $\sqrt{a}(0,1)$, and $w = \sqrt{a+b}$. Any pair of their compact group translates has Euclidean separation at least $\sqrt{2a}$. Quotient length distance dominates Euclidean orbit separation. Hence

$$\sqrt{2a} \le \operatorname{diam}_{d_\zeta} C_0(a,b) \le \frac{\pi}{\sqrt{2}}\sqrt{a}. \tag{64}$$

Interchanging vertices gives the corresponding bounds with b for C_2. Combined with (62), this proves the ratio-independent bounds of Theorem 1.1.

Suppose $a/b \to 0$ and $\rho \to 0$. After dividing by $\sqrt{b}$, put $\epsilon = a/b$. The entire plane satisfies $|z|^2 + |\varphi|^2 = \epsilon$ and $w = \sqrt{1+|\varphi|^2} = 1 + O(\epsilon)$. It approaches the wall point $w=1$ uniformly. Its stabilizer separates vertex 0 from the spherical pair $(1,2)$. The full normal representation has positive coordinates (z,φ_1,φ_2), their negative duals, and effective level ϵ. There is no tangent factor. A second rescaling by $\sqrt{\epsilon}$ gives the unit-level Calabi quotient on $T^*\mathbb{P}^2$, and the whole plane converges to its zero section. The unitary isometries of the Calabi quotient act transitively on this zero section. Thus every sequence of basepoints on P_0 has the same pointed limit up to isometry; no choice of a limiting point on the zero section changes the conclusion.

In that complete quotient

$$\{(q,p) : p(q) = 0,\ |q|^2 - |p|^2 = 1\}/\mathrm{U}(1),$$

the map $[q,p] \mapsto [q]$ is a retraction to the zero section. Since $|q| \ge 1$, its Fubini–Study differential is bounded by the ambient flat norm. Infimizing over vertical directions makes it 1-Lipschitz on the quotient. Ambient and intrinsic zero-section distances therefore agree, and their diameter is $\pi/2$. The relative comparisons at both scales prove

$$\frac{\operatorname{diam}_{g_\eta} P_0}{\sqrt{a}} \longrightarrow \frac{\pi}{2} \quad (a/b \to 0), \qquad \frac{\operatorname{diam}_{g_\eta} P_2}{\sqrt{b}} \longrightarrow \frac{\pi}{2} \quad (b/a \to 0). \tag{65}$$

8.5. **Successive complete limits.** The two-scale calculation is a particular case of a statement with moving centers. The necessary data include both the residual stability level and the normal basepoint.

Theorem 8.1. *Start with the actual rescaling (59)–(60). Let the current finite levels ν_n converge to ν_* and the current basepoints converge to $[x_*] \in \mathcal{Q}_{\nu_*}$. In the full normal slice at x_* write the representation as $W = U \oplus W'$ with compact stabilizer H, where $U = W^H$ and $W' = U^\perp$ for the invariant hyperkähler inner product. Write the balanced coordinates of the basepoints as $(u_n, z_n) \to 0$. Suppose $\sigma_n \downarrow 0$ and*

$$\frac{(\nu_n - \nu_*)|_{\operatorname{Lie} H}}{\sigma_n^2} \longrightarrow \lambda, \qquad \frac{z_n}{\sigma_n} \longrightarrow z_* \tag{66}$$

after compact group adjustment. At physical scale $\delta_n \sigma_n$ the complete pointed limit is

$$U \times (W' /\!/\!/_\lambda H), \qquad (0, [z_*]). \tag{67}$$

Here $/\!/\!/$ denotes the hyperkähler quotient with complex moment zero and real moment λ. The statement may be iterated through any fixed finite succession satisfying (66) at each stage. The physical scale is the product of the successive scales. No bound on $|u_n|/\sigma_n$ is required.

Proof. The full normal-slice and moving-center theorems [Wue26a, Sections 4 and 6] give the comparison after the change $(u, z) \mapsto (u_n + \sigma_n u, \sigma_n z)$. On bounded normal balls its relative squared-metric error is

$$O_{x_*}(|\nu_n - \nu_*| + |u_n| + \sigma_n R) = o(1). \tag{68}$$

Translation removes the tangential motion and does not change a moment level. The outer error (61) is also multiplicative, so it adds to this error and is not amplified by dividing by σ_n.

Choose a fixed normal sublevel whose twice-sized neighborhood lies in the slice. Its exit cost, divided by σ_n, tends to infinity. An actual path of length $O(\delta_n \sigma_n)$ therefore remains in both the outer chart and the inner slice; its translated normal radius is $O(\sigma_n)$. Conversely every bounded normal-model near-minimizer maps into these charts. Taking both length infima proves relative distance comparison, and the common inverse gives full ball coverage.

The existence of a module-simple point passes to every full normal Ext representation [Wue26a, Section 6]. Its finite metric theorem applies to the full $U \oplus W'$; the product decomposition then gives the complete target (67). For finitely many stages the relative errors sum to $o(1)$, and the successive exit estimates and inverse charts give the asserted iteration. Each limiting center has its own fixed radius and constants. The statement requires the convergent data at every stage and makes no assertion about an infinite succession. □

For this quiver the nonzero ζ_0 and ζ_2 walls have the isolated $T^*\mathbb{P}^2$ normal model. On a nonzero ζ_1 wall the tangent factor is $\mathbb{C}^2$ and the transverse model is $T^*\mathbb{P}^1$. If the internal scale in that pair-moduli surface also vanishes, the same full normal iteration applies. This describes approaches from all six chambers without a lower bound on a ratio of positive scales.

8.6. **The isolated and moving fibre constants.** In the isolated example the model plane is the zero section of $T^*\mathbb{P}^2$ at level κ_t. Its induced metric is exactly $\kappa_t g_{\mathrm{FS}}$, with line area $\pi\kappa_t$. The actual comparison has error $O(|t|)$ on that entire section. The first-exit argument and the 1-Lipschitz retraction give

$$\operatorname{diam}_{g_t} P_\pm = \frac{\pi}{2}\sqrt{\kappa_t}(1 + O(|t|)). \tag{69}$$

Dividing distances by $\sqrt{\kappa_t}$ gives the complete unit Calabi quotient, with the zero section as the limit of the actual plane.

For a moving fibre F_y over $y \in S^\circ$, the same argument in the two-arrow model gives

$$\operatorname{Area}(F_y) = \pi|\zeta_1|, \qquad \frac{\operatorname{diam}_{g_\eta} F_y}{\sqrt{|\zeta_1|}} \longrightarrow \frac{\pi}{2}. \tag{70}$$

The limit is through nonzero transverse levels as $\eta \to 0$, uniformly for y in a compact subset of S°. The complete normal limit is Eguchi–Hanson times the harmonic tangent plane. When $y \to o$, the fixed three-factor chart and the retained rescaled basepoint supply the applicable limit instead. These are the same fibres that specialize in (29).

9. The compact harmonic limit

The local estimates identify the geometry near the exceptional cycles. We finish by identifying the completion of the regular harmonic metric and the limit of the entire compact spaces. In the isolated example a direct collapsing-neighborhood argument suffices. For the interacting family we identify the limiting semiample class and apply the Ricci-flat degeneration theorems of Tosatti and Song to the actual harmonic metrics.

9.1. **Smooth convergence on the stable open set.** Let $U = Z_{\mathrm{reg}}$, identified with the H-stable bundle locus, and write g_0 for its central harmonic metric. A compact subset of U is covered by finitely many local universal families. At each central bundle the normalized fixed-determinant HE equation has invertible trace-free gauge linearization, since stability removes its kernel. Compactness supplies a uniform positive lower bound. The parameter implicit-function theorem and elliptic bootstrapping give smooth variation of the normalized HE metrics. The deformation kernels have constant dimension on this smooth locus, so their harmonic projections vary smoothly as well. Consequently

$$g_\eta \longrightarrow g_0 \quad \text{in every fixed } C^k \text{ norm on compact subsets of } U. \tag{71}$$

These are the same holomorphic bundles in every nearby chamber. The identification is the restriction of the classifying contraction.

9.2. **A direct argument for the isolated plane.** In the isolated zero-level model, distance to the cone vertex is the norm radius. The local harmonic comparison shows that a radius-ρ neighborhood has diameter $O(\rho)$ and that adding the vertex is exactly the local metric completion, with its quotient topology. Outside such a neighborhood the target is compact and smooth. Thus the completion of (U, g_0) is Z, with no further end.

One can also see compact convergence directly. Use the invariant $A = u \otimes \varphi$ from (21), and take the target neighborhood where its nonzero singular value is at most $\rho^2/2$. In the balanced model, if $\kappa_t \le \rho^2$, its whole preimage has radius $O(\rho)$ and diameter $O(\rho)$, uniformly in t. Whole-fibre coverage and the ambient comparison transfer this bound to the actual moduli space.

Collapse each of these neighborhoods to a point. A shortest quotient path needs to visit that point at most once; lifting the visit across a set of diameter $O(\rho)$ changes distance by at most $O(\rho)$. For fixed ρ, the common compact exterior has smoothly convergent metrics by (71), so the collapsed spaces converge in length distance. The limsup of the original compact Gromov–Hausdorff distance is therefore at most $C\rho$. Letting $\rho \downarrow 0$ proves convergence to the central completion.

9.3. **The limiting class gives the actual contraction.** In the interacting example put $c = D_1 + D_2$ and $b_0 = (0, H, 0) \in v^\perp$. On every chamber,

$$\frac{[\Omega_\eta]}{2\pi^2} \longrightarrow \theta_v(b_0).$$

To identify this limiting class globally, use the Greb–Toma determinant construction with a sufficiently large very ample multiple mH. Its surface K-class is

$$u_1 = -3[\mathcal{O}_{mH}] + \chi(E \otimes \mathcal{O}_{mH})[\mathcal{O}_x].$$

Riemann–Roch gives

$$v(\mathcal{O}_{mH}) = (0, mH, -(mH)^2/2), \qquad v(u_1) = (0, -3mH, c \cdot mH).$$

Their determinant convention is $\lambda_{\mathcal{E}}(u_1) = \det Rp_*(\mathcal{E} \otimes p_X^* u_1)$ [GT17, Definition 2.1]. Since our Mukai convention inserts the dual vector, and $c \cdot H = 0$, we obtain

$$c_1 \lambda_{\mathcal{E}}(u_1) = 3m\theta_v(b_0). \tag{72}$$

The classifying theorem gives an ample line on the slope target whose pullback is $\lambda_{\mathcal{E}}(u_1)^N$ for a positive integer N. Restrict to the common reduced image and pull back under its finite normalization. The resulting line A_Z on Z is ample. Using a common power for the finitely many chambers gives

$$\theta_v(b_0) = \pi_\sigma^* \alpha_Z, \qquad \alpha_Z = \frac{c_1(A_Z)}{3mN}. \tag{73}$$

The maps here are precisely the classifying contractions already constructed. Thus the limiting class is big and nef, and its sufficiently divisible multiples define this connected-fibre semiample morphism. Vanishing on the named exceptional curves alone would not have identified the ample class on the target.

9.4. **Ricci-flatness and metric identification.** Every chamber object is locally free and slope stable. At fixed determinant its trace-free HE connection is an irreducible projective ASD connection on the Ricci-flat K3 surface. Itoh's construction [Ito88, Theorem 2.1] makes its smooth moduli metric hyperkähler. The real Dolbeault metric differs by the fixed factor specified in Section 2; it is therefore hyperkähler as well, smooth on the entire compact M_σ, and complete. By uniqueness in a Kähler class, $g_\eta/(2\pi^2)$ is the Ricci-flat metric in class $\theta_v(b_\eta)$ on that fixed complex manifold. No auxiliary metric on a deformation-equivalent space is used.

The symplectic form on U pulls back and extends as the symplectic form on M_σ. Its top power trivializes the canonical sheaf on U. Normality and the codimension-at-least-two singular complement give the trivial reflexive canonical sheaf on Z. The resolution has zero discrepancies; further resolutions have nonnegative discrepancies, since M_σ is smooth. Hence Z is a projective Calabi–Yau variety with canonical singularities and a crepant resolution, as required by the metric completion results.

Tosatti's real-class degeneration theorem [Tos09, Theorem 1.1 and the subsequent sequence formulation] applies to $\theta_v(b_\eta) \to \pi_\sigma^* \alpha_Z$. These classes lie in the real Néron–Severi space. The null locus of the limiting semiample class is precisely the exceptional locus of its contraction π_σ. Its smooth limit away from the exceptional locus is the regular Ricci-flat metric of α_Z on Z. Equation (71) identifies that metric with $g_0/(2\pi^2)$. This equality of metric tensors is a separate step from the algebraic/Uhlenbeck topological identification.

Proposition 9.1. *The completion of (U, g_0) is a compact length space (Z, d_0) whose topology is the projective topology. For every generic sequence $\eta \to 0$, allowing changes of chamber and arbitrary ratios of its positive scales,*

$$(M_{\sigma(\eta)}, g_\eta) \longrightarrow (Z, d_0) \quad \textit{in compact Gromov–Hausdorff distance.}$$

Proof. Song's completion theorem [Son14, Theorem 1.1] applies to (Z, α_Z). Its regular Ricci-flat metric has compact completion homeomorphic to Z, extending the identity on U. The metric identification above and a fixed homothety give the first assertion. On a fixed chamber, the limiting class is rational, big and nef, with semiample model exactly Z by (73). Given any sequence of interior classes tending to it, join consecutive classes by segments in the convex Kähler cone of M_σ. Parametrize these segments on intervals accumulating at zero, smoothly at the joining times. The resulting family stays Kähler for positive parameter, passes through the prescribed classes and tends to the same boundary class. Song's Corollary 1.1 applies to its Ricci-flat metrics and hence to the original sequence. Only the limiting class is required to be rational; real interior classes are allowed.

There are only six chambers. Every generic parameter sequence has subsequences in a single chamber. The determinant calculation and the regular harmonic identification give the same (Z, d_0) for all such subsequences. The subsequence criterion proves the assertion for the original sequence. □

Proof of Theorem 1.1. Section 4 constructs the compact chamber spaces, excludes all ideal charge and identifies the full singular locus of their common normal target. Section 5 gives all six deepest fibres and the reduced specialization. The primitive cycle calculations and Proposition 6.1 prove the integral assertions. Equations (54), (62), (64) and (65), together with Theorem 8.1, give the exact periods, ambient collapse and complete rescaled limits for those same cycles. Proposition 9.1 proves the final compact convergence. □

The integral attachment uses a marked complement, and the metric completion uses the global harmonic class on the actual compact family. The exceptional cycles thus carry explicit global information, while the geometry retained away from them remains part of the reconstruction data. The argument concerns these contractions without lost charge. Extending it to singular graded sheaves and bubbling would require compatible local metrics and integral attachment data for those additional boundary strata.

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